ANNAMALAI UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

B.E. (COMPUTER SCIENCE AND ENGINEERING)

SEMESTER – VII

22CSCP706 – EMBEDDED SYSTEMS AND INTERNET OF THINGS **(IOT) LAB**

Name :...

Reg. No :..

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B.E CSE (COMPUTER SCIENCE AND ENGINEERING)

SEMESTER – VII

22CSPC706 – EMBEDDED SYSTEMS AND INTERNET OF THINGS (IoT) LAB

Bonafide Certificate

Certified that this is the bonafide record of work done by Mr./Ms.____ Reg. No. ________________ of B.E. Computer Science and Engineering in the 22CSPC706 – EMBEDDED SYSTEMS AND INTERNET OF THINGS (IOT) LAB during the odd semester of the academic year 2025-2026.

Staff-in-charge   Internal Examiner

Place : Annamalai Nagar   External Examiner

Date :

Vision and Mission of the Department

VISION

To provide a congenial ambience for individuals to develop and blossom as academically superior, socially conscious and nationally responsible citizens.

MISION

M1: Impart high quality computer knowledge to the students through a dynamic scholastic environment wherein they learn to develop technical, communication and leadership skills to bloom as a versatile professional.

M2: Develop life-long learning ability that allows them to be adaptive and responsive to the changes in career, society, technology, and environment.

M3: Build student community with high ethical standards to undertake innovative research and development in thrust areas of national and international needs.

M4: Expose the students to the emerging technological advancements for meeting the demands of the industry.

Program Educational Objectives (PEOs)

PEOs	**PEO Statements**
PEO1	To prepare graduates with potential to get employed in the right role and/or become entrepreneurs to contribute to the society.
PEO2	To provide the graduates with the requisite knowledge to pursue higher education and carry out research in the field of Computer Science.
PEO3	To equip the graduates with the skills required tostay motivated and adapt to the dynamically changing world so as to remain successful in their career.
PEO4	To train the graduates with effectively, work collaboratively and exhibit high levels of professionalism and ethical responsibility.

COURSE OUTCOMES:

At the end of this course, the students will be able to

1. Develop a real time projects using embedded systems including 8051 and Advanced RISC Machines (ARM).
2. Design IoT based products that can be used in all real time applications.
3. Demonstrate an ability to listen and answer the viva questions related to programming

skills needed for solving real-world problems in Computer Science and Engineering.

Mapping of Course Outcomes with Programme Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	2	-	2	-	-	-	-	-	-	-
CO2	-	3	3	1	3	1	-	-	-	-	-	2
CO3	2	2	-	-	-	-	-	-	-	2	-	2

Rubric for CO3

Rubric for CO3 in Laboratory Courses				
Distribution of 10 Marks for CIE/SEE Evaluation Out of 40/60 Marks				
Rubric	Up To 2.5 Marks	Up To 5 Marks	Up To 7.5 Marks	Up To 10 marks
Demonstrate an ability to listen and answer the viva questions related to programming skills needed for solving real-world problems in Computer Science and Engineering.	Poor listening and communication skills. Failed to relate the programming skills needed for solving the problem.	Showed better communication skill by relating the problem with the programming skills acquired but the description showed serious errors.	Demonstrated good communication skills by relating the problem with the programming skills acquired with few errors.	Demonstrated excellent communication skills by relating the problem with the programming skills acquired and have been successful in tailoring the description.

INDEX

INTERNET OF THINGS

Introduction:

Internet is a network of live persons. Internet of Things or IoT is a network of things and persons. IoT brings the things alive and there is interacting among themselves and with persons lively. Internet of Things is the network of devices such as vehicles, and home appliances that contain electronics, software, actuators, and connectivity which allows these things to connect, interact and exchange data. With the help of embedded technology, these things can communicate and interact over the Internet, and they can be remotely monitored and controlled. In the Internet of things, the precise geographic location and also the dimensions of a thing is critical. Therefore, sensors, transducers, locating devices and networks play very important role in IoT.

IoT makes virtually everything "smart," by improving aspects of our life with the power of data collection, AI algorithm, and networks. The thing in IoT can also be a person with a diabetes monitor implant, an animal with tracking devices, etc. There are many areas of applications of the Internet of Things like consumer, health, industrial, transportation, security, entertainment and many other. Security, Privacy, Complexity, Compliance, are key challenges of IoT

Components of IOT:

- **Sensors/Devices:** Sensors or devices are a key component that helps you to collect live data from the surrounding environment.
- **Connectivity:** All the collected data is sent to a cloud infrastructure. The sensors should be connected to the cloud using various mediums of communications like Bluetooth, WI-FI, WAN, etc.
- **Data Processing:** Once that data is collected, and it gets to the cloud, the software performs processing on the gathered data.
- **User Interface:** The information needs to be available to the end-user in some way which can be achieved by triggering alarms on their phones or sending them notification through email or text message.

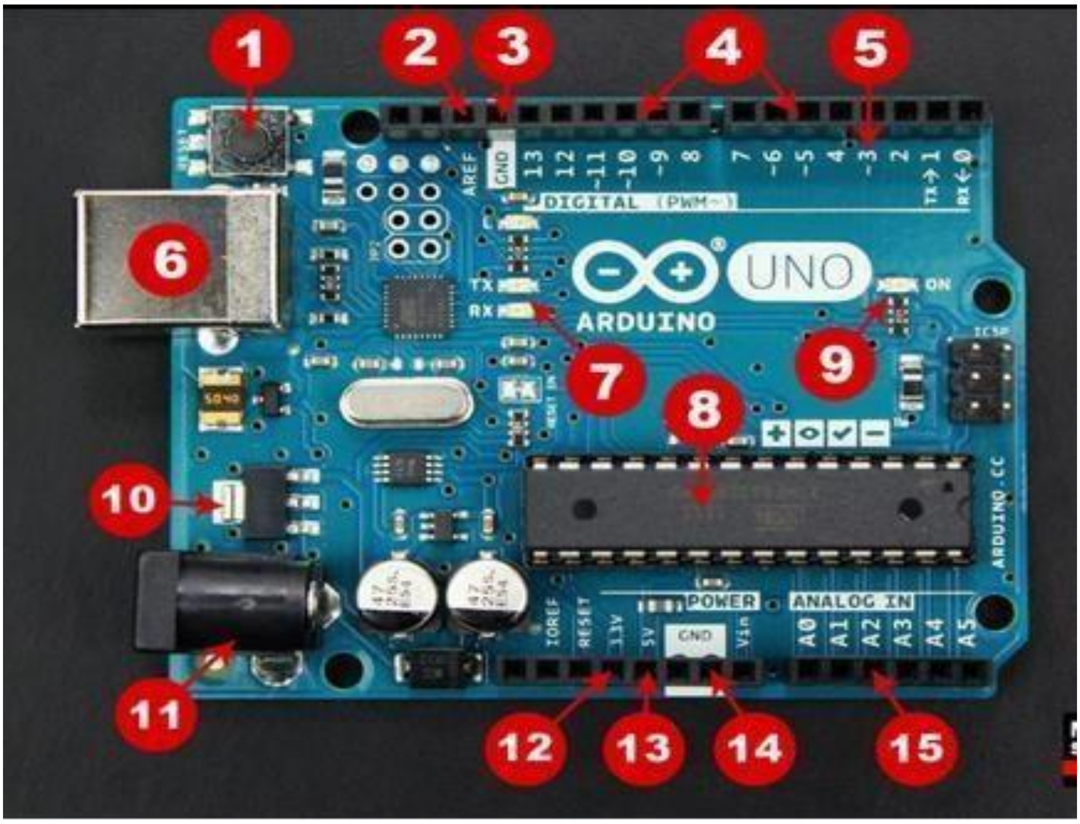
Advantages of IOT:

- Technical optimization
- Improved data collection
- Reduced waste
- Improved customer engagement

Arduino UNO

Introduction:

Arduino is an open source programmable circuit board that can be integrated into a wide variety of makerspace projects both simple and complex. This board contains a microcontroller which is able to be programmed to sense and control objects in the physical world. By responding to sensors and inputs, the Arduino is able to interact with a large array of outputs such as LEDs, motors and displays. Because of its flexibility and low cost, Arduino has become a very popular choice for makers and makerspaces looking to create interactive hardware projects.



Here are the components that make up an Arduino board and what each of their functions are.

1. **Reset Button** – This will restart any code that is loaded to the Arduino board

2. **AREF** – Stands for “Analog Reference” and is used to set an external reference voltage

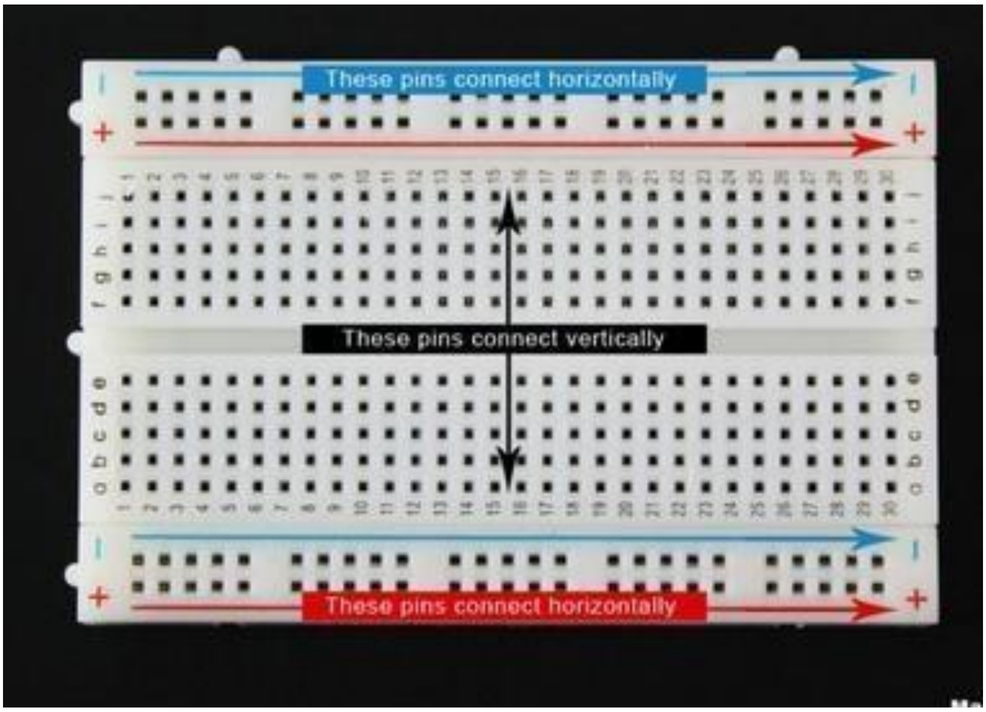
3. **Ground Pin** – There are a few ground pins on the Arduino and they all work the same
4. **Digital Input/Output** – Pins 0-13 can be used for digital input or output
5. **PWM** – The pins marked with the (~) symbol can simulate analog output
6. **USB Connection** – Used for powering up your Arduino and uploading sketches
7. **TX/RX** – Transmit and receive data indication LEDs
8. **ATmega Microcontroller** – This is the brains and is where the programs are stored
9. **Power LED Indicator** – This LED lights up anytime the board is plugged in a power source
10. **Voltage Regulator** – This controls the amount of voltage going into the Arduino board
11. **DC Power Barrel Jack** – This is used for powering your Arduino with a power supply
12. **3.3V Pin** – This pin supplies 3.3 volts of power to your projects
13. **5V Pin** – This pin supplies 5 volts of power to your projects
14. **Ground Pins** – There are a few ground pins on the Arduino and they all work the same
15. **Analog Pins** – These pins can read the signal from an analog sensor and convert it to digital

Power supply:

The Arduino Uno needs a power source in order for it to operate and can be powered in a variety of ways. The most common way is to connect the board directly to the computer via a USB cable.

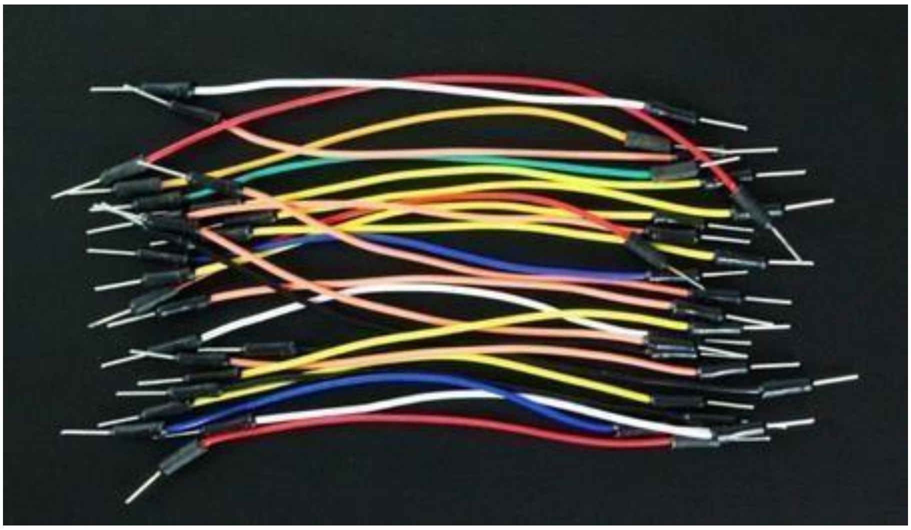
Arduino Breadboard

This device allows you to prototype your Arduino project without having to permanently solder the circuit together. Using a breadboard allows you to create temporary prototypes and experiment with different circuit designs. Inside the holes (tie points) of the plastic housing, are metal clips which are connected to each other by strips of conductive material.

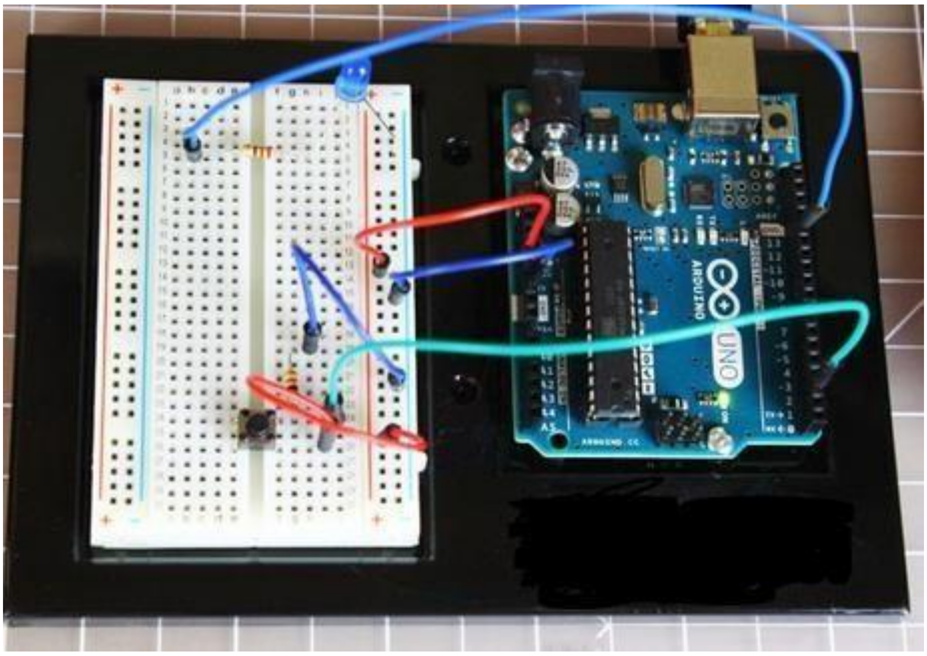


Jumper wire:

The breadboard is not powered on its own and needs power brought to it from the Arduino board using jumper wires. These wires are also used to form the circuit by connecting resistors, switches and other components together.



Here is a visual of what a completed Arduino circuit looks like when connected to a breadboard.



Once the circuit has been created on the breadboard, program (known as a sketch) need to be uploaded to the Arduino. The sketch is a set of instructions that tells the board what functions it needs to perform. An Arduino board can only hold and perform one sketch at a time. The software used to create

Arduino sketches is called the IDE which stands for Integrated Development Environment. The software is free to download and can be found at https://www.arduino.cc/en/Main/Software

Every Arduino sketch has two main parts to the program:

void setup() – Sets things up that have to be done once and then don't happen again.

void loop() – Contains the instructions that get repeated over and over until the board is turned off.

EX. NO: 01	DISTANCE MEASUREMENT USING ARDUINO
DATE :	

Aim:

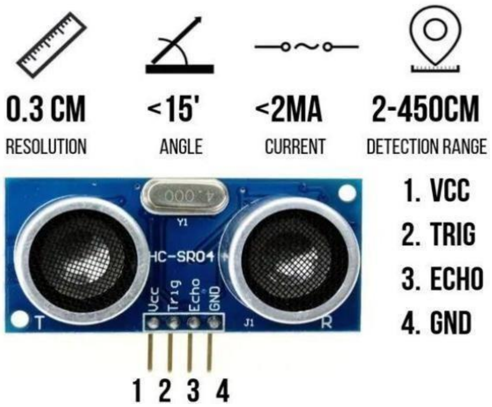
To measure the distance of an object using ultrasonic sensor HC–SR04.

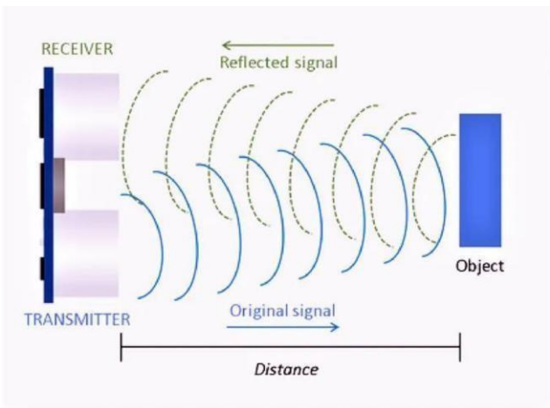
List of components:

1. Solderless Breadboard Full size
2. Jumper Wires
3. Arduino UNO
4. Buzzer
5. LEDs
6. Ultrasonic Sensor HC-SR04

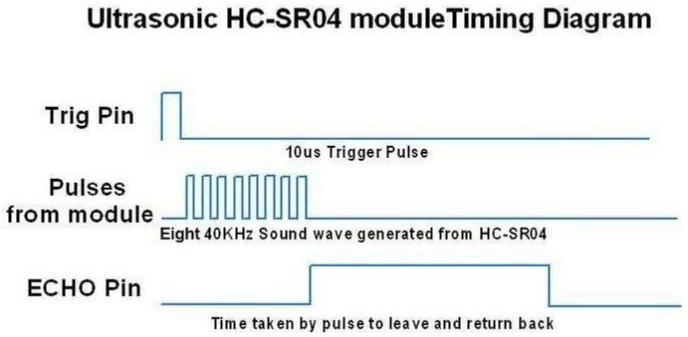
Working description:

Ultrasonic Sensor HC-SR04 is a sensor that can measure distance. It emits an ultrasound at 40 000 Hz (40 kHz) which travels through the air and if there is an object or obstacle on its path It will bounce back to the module. Considering the travel time and the speed of the sound you can calculate the distance. The configuration pin of HC-SR04 is VCC (1), TRIG (2), ECHO (3), and GND (4). The supply voltage of VCC is +5V and you can attach TRIG and ECHO pin to any Digital I/O in your Arduino Board.

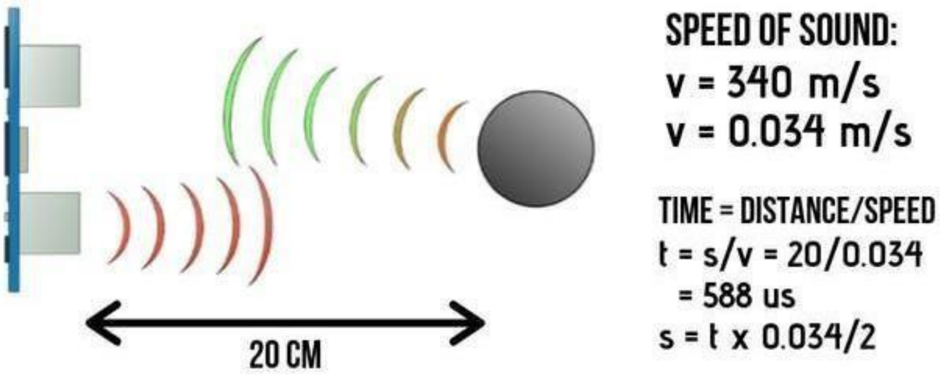




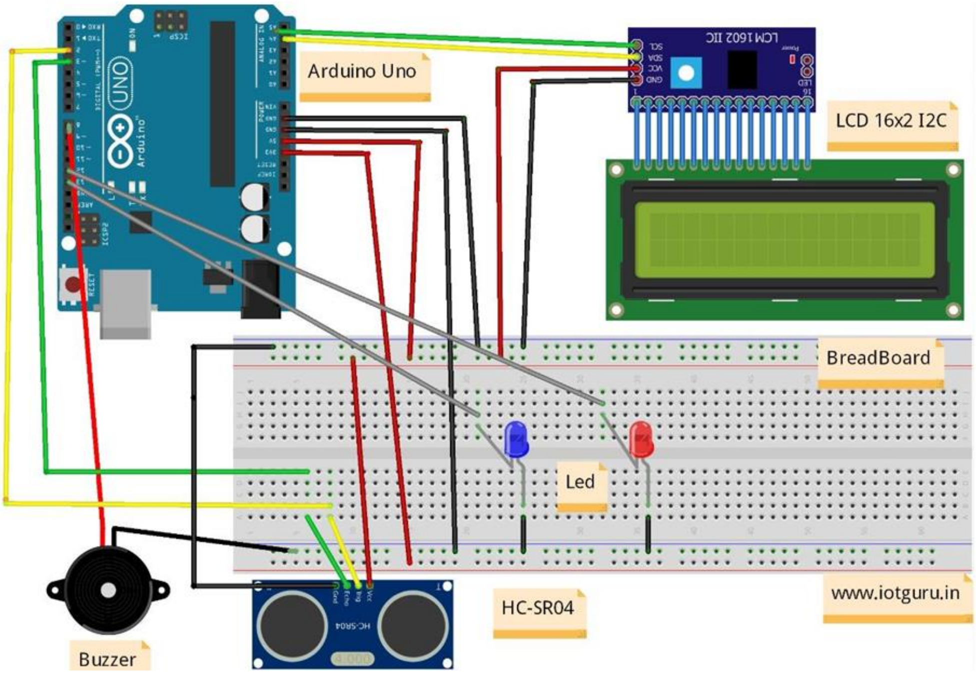
In order to generate the ultrasound we need to set the Trigger Pin on a High State for 10 µs. That will send out an 8 cycle sonic burst which will travel at the speed sound and it will be received in the Echo Pin. The Echo Pin will output the time in microseconds the sound wave travelled.



For example, if the object is 20 cm away from the sensor, and the speed of the sound is 340 m/s or 0.034 cm/µs the sound wave will need to travel about 588 microseconds. But what you will get from the Echo pin will be double that number because the sound wave needs to travel forward and bounce backward. So in order to get the distance in cm we need to multiply the received travel time value from the echo pin by 0.034 and divide it by 2.



Circuit diagram:



Connections:

Ultrasonic sensor to Arduino board:

Vcc to 5v

Trig(input) to pin 2

Gnd to gnd

Echo (output) to pin 3

LED BLUE to Arduino board:

Negative to gnd

Positive to pin 13

LED RED to Arduino board:

Negative to gnd

Positive to pin 12

Buzzer to Arduino board:

Negative to gnd

Positive to pin 8

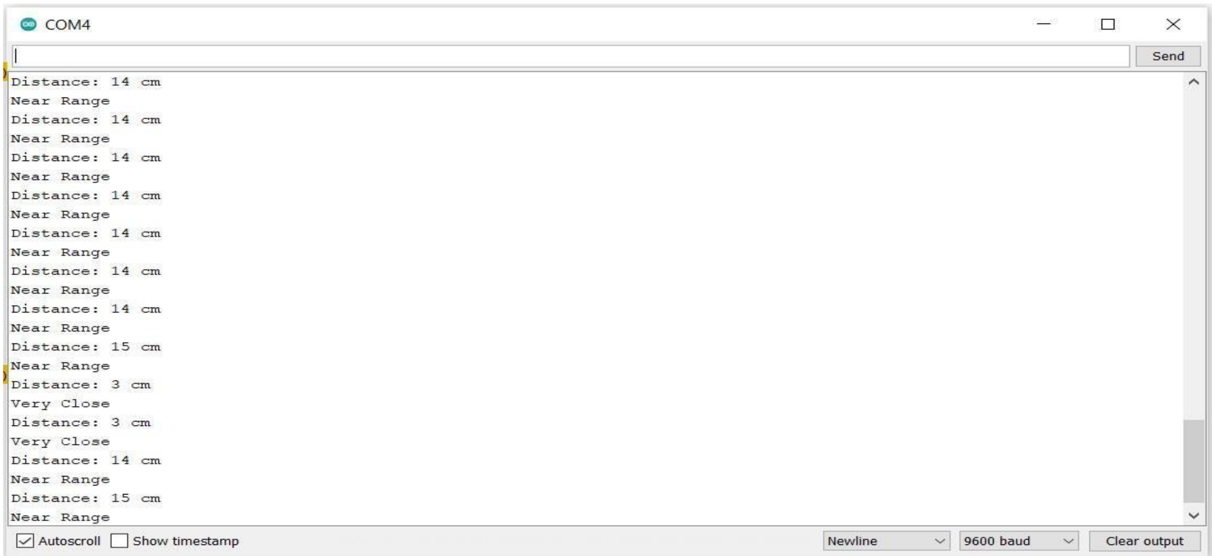
Code:

```
#include <Wire.h>
#define echoPin 3
#define trigPin 2
long duration;
int distance;
const int buzzer = 8;
const int light = 13;
const int light2 = 12;
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
  pinMode(buzzer, OUTPUT);
  pinMode(light, OUTPUT);
  Serial.println("Annamalai University BE CSE 2022 "); // print some text in
Serial Monitor
  Serial.println("Distance Measure Program");
}
```

```
void loop()
 { digitalWrite(trigPin,
 LOW);
 delayMicroseconds(2);
 digitalWrite(trigPin, HIGH);
 delayMicroseconds(10);
 digitalWrite(trigPin, LOW);
 duration = pulseIn(echoPin, HIGH);
 distance = duration * 0.034 / 2;
 Serial.print("Distance: ");
 Serial.print(distance);
 Serial.println(" cm");
 digitalWrite(light, LOW);
 digitalWrite(light2, LOW);
 if (distance <= 10)
 {
     Serial.println("Very Close");
  digitalWrite(light, HIGH);
  tone(buzzer, 1000); // Send 1KHz sound signal...
  delay(1000);        // ...for 1 sec
  digitalWrite(light2, LOW);
  tone(buzzer, 500); noTone(buzzer);
                              // Stop sound...
 }
 else if (distance >=11 && distance <=50)
```

```
{
  Serial.println("Near Range");
  digitalWrite(light2, HIGH);
  digitalWrite(light, LOW);
  delay(1000);
}
else
{
  Serial.println("Far Range");
}}
```

Output:



Result:

Thus the distance of an object was measured using Arduino successfully.

EX. NO:02	IDENTIFYING MOISTURE CONTENT IN AGRICULTURAL LAND USING ARDUINO
DATE :	

Aim:

To identify the moisture content of soil in Agricultural Land using Soil Moisture Sensor.

List of components:

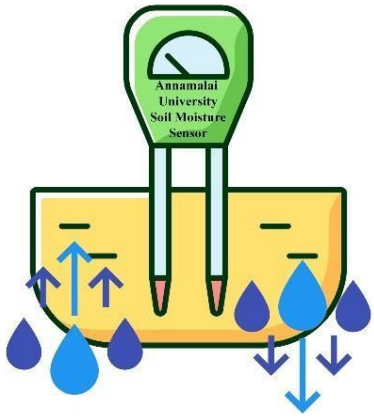
1. Solderless Breadboard Full Size
2. Jumper Wires
3. Arduino UNO
4. Buzzer
5. LEDs
6. Soil Moisture sensor

Working Description

Soil moisture sensors measure the volumetric water content in soil. Since the direct gravimetric measurement of free-soil moisture requires removing, drying, and weighing of a sample, soil moisture sensors measure the volumetric water content indirectly by using some other property of the soil, such as electrical resistance, dielectric constant, or interaction with neutrons, as a proxy for the moisture content.

The relation between the measured property and soil moisture must be calibrated and may vary depending on environmental factors such as soil type, temperature, or electric conductivity. Reflected microwave radiation is affected by the soil moisture and is used for remote sensing in hydrology and agriculture. Portable probe instruments can be used by farmers or gardeners.

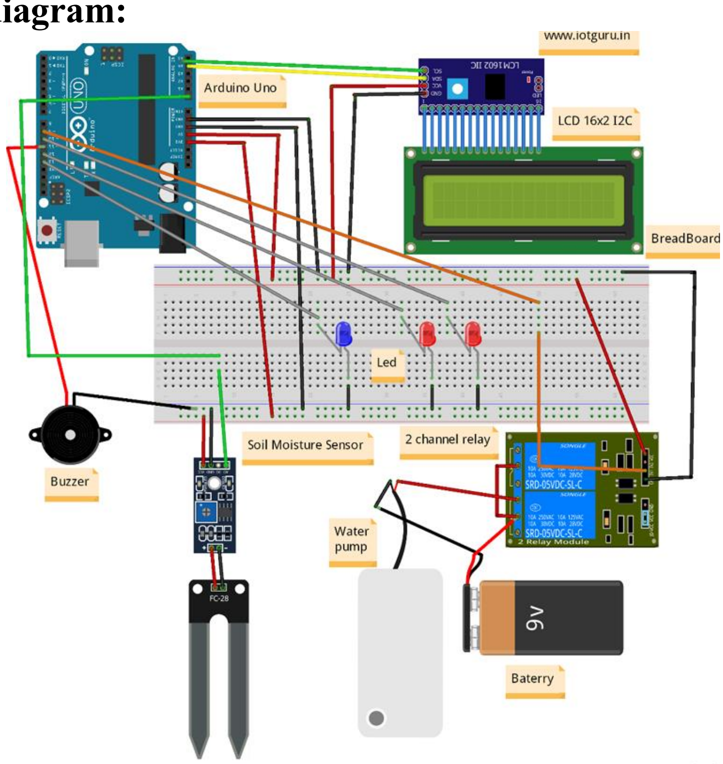
Soil moisture sensors typically refer to sensors that estimate volumetric water content. Another class of sensors measure another property of moisture in soils called water potential; these sensors are usually referred to as soil water potential sensors and include tensiometers and gypsum blocks.



There are different types of soil moisture sensor on the market, but their working principal are all similar. All of these sensors have at least three pins: VCC, GND, and AO. The AO pin changes according to the amount of moisture in the soil and increases as there is more water in the soil. Some models have an additional base called DO. If the moisture amount is less than the permissible amount (which can be changed by the potentiometer on the sensor) the DO pin will be “1”, otherwise will remain”0″.

In this Experiment, we have used the Soil Moisture Sensor. It has a detection length of 38mm and a working voltage of 2V-5V. It has a Fork-like design, which makes it easy to insert into the soil. The analog output voltage boosts along with the soil moisture level increases.

Circuit diagram:



Connections:

Soil moisture sensor to Arduino board:

AO to A0

Vcc to 5V

Gnd to gnd of buzzer

Buzzer to Arduino board:

+ve leg to pin 1

LED1 to Arduino board:

Gnd to gnd

+ve leg to pin 13

LED2 to Arduino board:

Gnd to gnd

+ve leg to pin 10

LED3 to Arduino board:

Gnd to gnd

+ve leg to pin 12

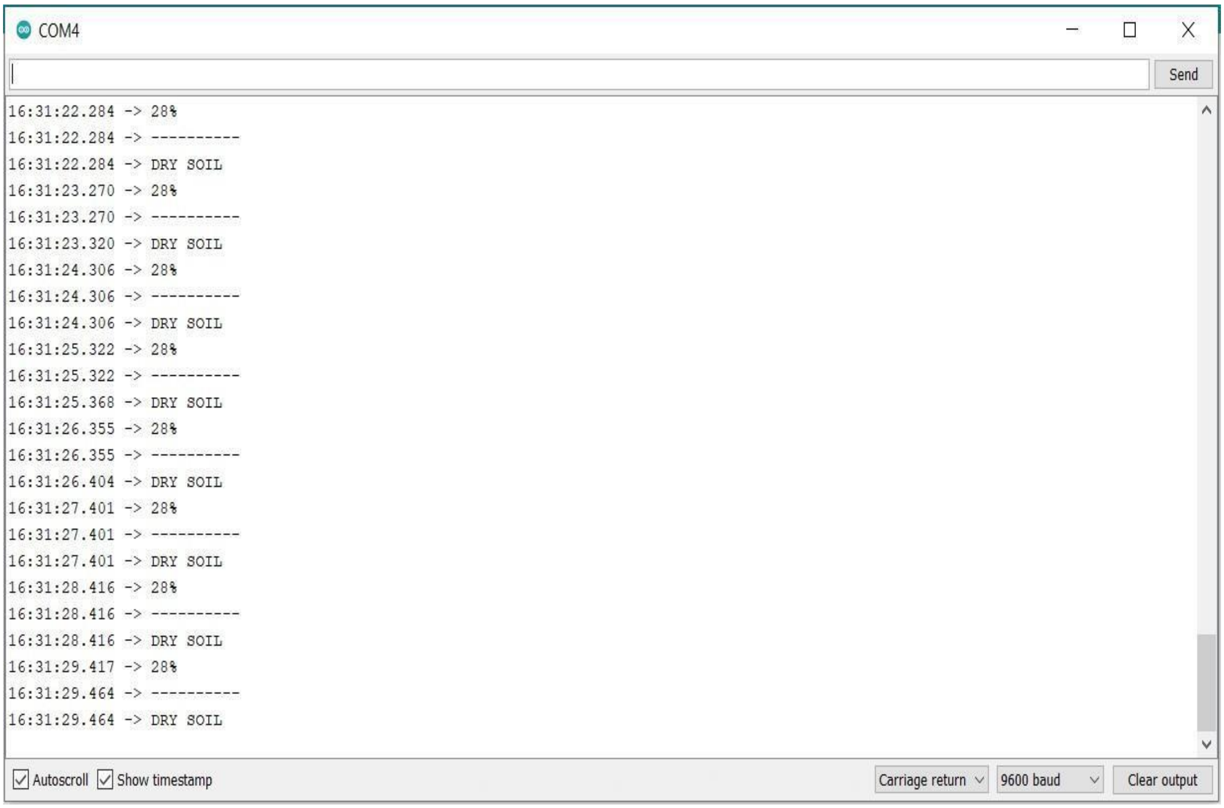
Code:

```
#include <Wire.h>

const int buzzer = 11;

const int light = 13;

const int light1 = 10;

const int light2 = 12;

int moisture;

void setup(){
 pinMode(A0, INPUT);
 Serial.begin(9600);

 pinMode(buzzer, OUTPUT);

 pinMode(light, OUTPUT);

 pinMode(light1, OUTPUT);

 pinMode(light2, OUTPUT);

 pinMode(9, OUTPUT);

 Serial.println("Soil Moisture

 Program");

 }

void loop() {

 int SensorValue = analogRead(A0);

  moisture = (100 - ((SensorValue/1023.00) * 100));

 Serial.print(moisture);

 Serial.println("%");

 Serial.println(" --------- ");

 if (SensorValue >= 1000)
```

```
{
 Serial.println("Not in soil or disconnected");
}
if (SensorValue < 1000 && SensorValue >= 600)
{
   digitalWrite(9, LOW);
   Serial.println("DRY SOIL");
   digitalWrite(light, HIGH);
   tone(buzzer, 1000);
   delay(1000);
   digitalWrite(light1, LOW);
   digitalWrite(light2, LOW);
   tone(buzzer, 500); noTone(buzzer);

}
if (SensorValue < 600 && SensorValue >= 370)
{
   digitalWrite(9, HIGH);
   Serial.println("HUMID SOIL");
   digitalWrite(light1, HIGH);
   digitalWrite(light2, LOW);
   digitalWrite(light, LOW);
   delay(1000);
}
if (SensorValue < 370)
{
   digitalWrite(9, HIGH);
```

```
    Serial.println("WATER SOIL");
    digitalWrite(light2, HIGH);
    digitalWrite(light, LOW);
    digitalWrite(light1, LOW);
    delay(1000);
} }
```

Output:



Result:

Thus the moisture content in Agricultural land was measured using Arduino successfully.

EX. NO: 03	MOTION DETECTION USING ARDUINO
DATE :	

Aim:

To build a motion detection system using Arduino.

List of components

1. Solderless Breadboard Full Size
2. Jumper Wires
3. Arduino UNO
4. Buzzer
5. LEDs
6. PIR Motion sensor

Working Description

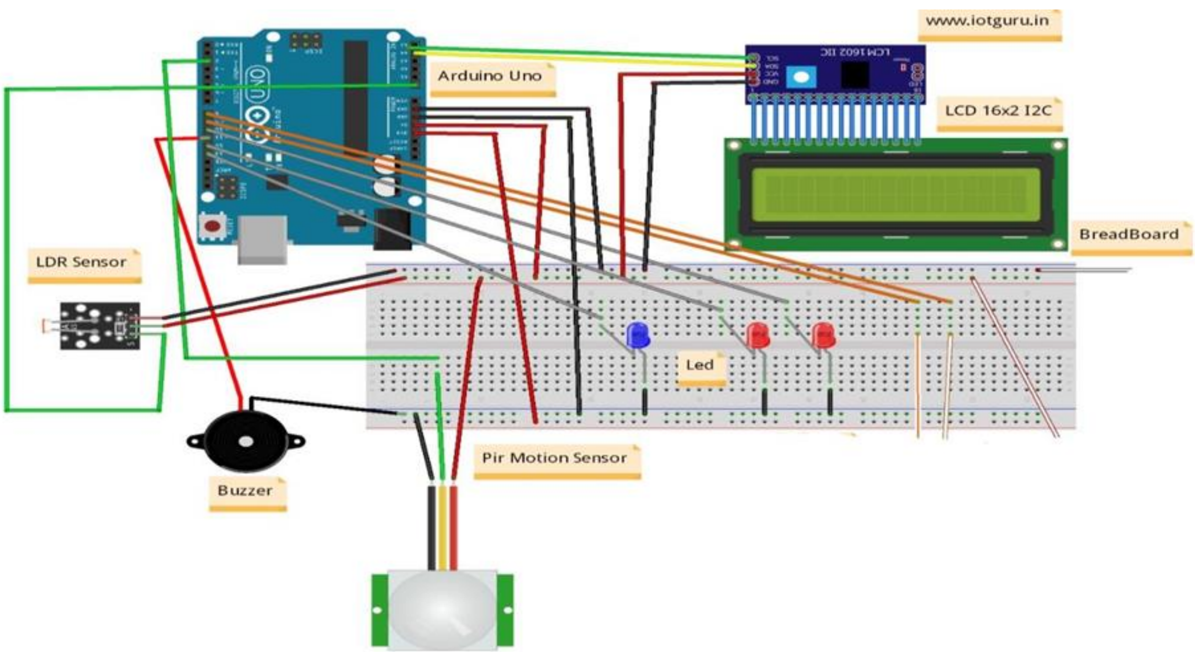
Home Automation is what makes us call our place a Smart Home. We can control all the appliances at our home using automation. The devices within the home automation system connect with each over a local network. When connected with the Internet, home devices are an important constituent of the Internet of Things. Mainly Home automation is to control or monitor signals from different appliances. Simply, this helps to make our lives easy.

An LDR (Light Dependent Resistor) is a component that has a (variable) resistance that changes with the light intensity that falls upon it. This allows them to be used in light sensing circuits. A photoresistor is made of a high resistance semiconductor. In the dark, a photoresistor can have a resistance as high as several megohms (MΩ), while in the light, a photoresistor can have a resistance as low as a few hundred ohms. If incident light on a photoresistor exceeds a certain frequency, photons absorbed by the semiconductor give bound electrons enough energy to jump into the conduction band.

The resulting free electrons (and their hole partners) conduct electricity, thereby lowering resistance. This example demonstrates the use of a LDR as a switch. Each time you cover the photocell, the LED (or whatever) is turned on or off.

PIR sensor detects a human being moving around within approximately 10m from the sensor. This is an average value, as the actual detection range is between 5m and 12m.PIR are fundamentally made of a pyro electric sensor, which can detect levels of infrared radiation. For numerous essential projects or items that need to discover when an individual has left or entered the area. PIR sensors are incredible, they are flat control and minimal effort, have a wide lens range, and are simple to interface with.

Circuit diagram:



Connections:

PIR sensor to Arduino board:

Gnd to gnd of buzzer

Vcc to 5V

Output to pin 2

LDR sensor to Arduino board:

Vcc to 5V

Gnd to gnd

Buzzer to Arduino board:

+ve pin to pin 11

LED1 to Arduino board:

Gnd to gnd

+ve leg to pin 13

LED2 to Arduino board:

Gnd to gnd

+ve leg to pin 12

LED3 to Arduino board:

Gnd to gnd

+ve leg to pin 10

Code:

```
#include <Wire.h>
const int buzzer = 11;
const int light = 13;
int sensor = 2;              // the pin that the sensor is atteched to
int state = LOW;             // by default, no motion detected
int val = 0;                 // variable to store the sensor status (value)
void setup() {
  Serial.begin(9600);
  pinMode(buzzer, OUTPUT);
  pinMode(light, OUTPUT);
  pinMode(sensor, INPUT);     // initialize sensor as an input
  Serial.println("Annamalai University BE CSE 2022 "); // print some text in
Serial Monitor
  Serial.println("Home Automation Program");
}
    void loop() {
    val = digitalRead(sensor);
    if (val == HIGH) {

    delay(100);
     if (state == LOW)
     { Serial.println("Motion
     Detected");Serial.println("");
     digitalWrite(light, HIGH);
     tone(buzzer, 1000);
     delay(500);
```

```
     tone(buzzer, 500);
      noTone(buzzer);
      state = HIGH;
    }
  }
  else {
      delay(100);

      if (state == HIGH){
        Serial.println("Motion stopped!");
        digitalWrite(light, LOW);
        state = LOW;
    }
  }
}
```

Output:

Motion not detected..

Motion detected..

Motion detected.. Motion

not detected..

Result:

Thus the Motion detection System was built successfully using Arduino.

EX. NO: 04	IDENTIFYING ROOM TEMPERATURE AND HUMIDITY USING ARDUINO
DATE :	

Aim:

To identify the room temperature and humidity using DHT11 sensor.

List of components

1. Solderless Breadboard Full Size
2. Jumper Wires
3. Arduino UNO
4. Buzzer
5. LEDs
6. DHT11 Sensor

Working Description

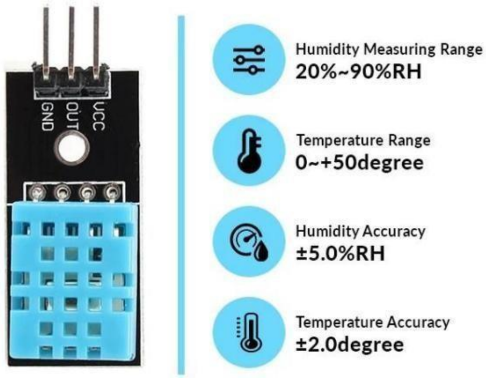
DHT11 is a low-cost digital sensor for sensing temperature and humidity. This sensor can be easily interfaced with any micro-controller such as Arduino, Raspberry Pi etc. to measure humidity and temperature instantaneously. DHT11 humidity and temperature sensor is available as a sensor and as a module. The difference between this sensor and module is the pull-up resistor and a power-on LED. DHT11 is a relative humidity sensor. To measure the surrounding air this sensor uses a thermistor and a capacitive humidity sensor.

Working Principle of DHT11 Sensor

DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature. The humidity sensing capacitor has two electrodes with a moisture holding substrate as a dielectric between them. Change in the capacitance value occurs with the change in humidity levels. The IC measure, process this changed resistance values and change them into digital form.

For measuring temperature this sensor uses a Negative Temperature coefficient thermistor, which causes a decrease in its resistance value with increase in temperature. To get larger resistance value even for the smallest change in temperature, this sensor is usually made up of semiconductor ceramics or polymers.

The temperature range of DHT11 is from 0 to 50 degree Celsius with a 2-degree accuracy. Humidity range of this sensor is from 20 to 80% with 5% accuracy. The sampling rate of this sensor is 1Hz .i.e. it gives one reading for every second. DHT11 is small in size with operating voltage from 3 to 5 volts. The maximum current used while measuring is 2.5mA.



Connections:

DHT 11 sensor to Arduino board:

Gnd to gnd

Vcc to 5V

Out to pin 2

LED1 to Arduino board:

Gnd to gnd

+ve leg to pin 5

LED2 to Arduino board:

Gnd to gnd

+ve leg to pin 6

LED3 to Arduino board:

Gnd to gnd

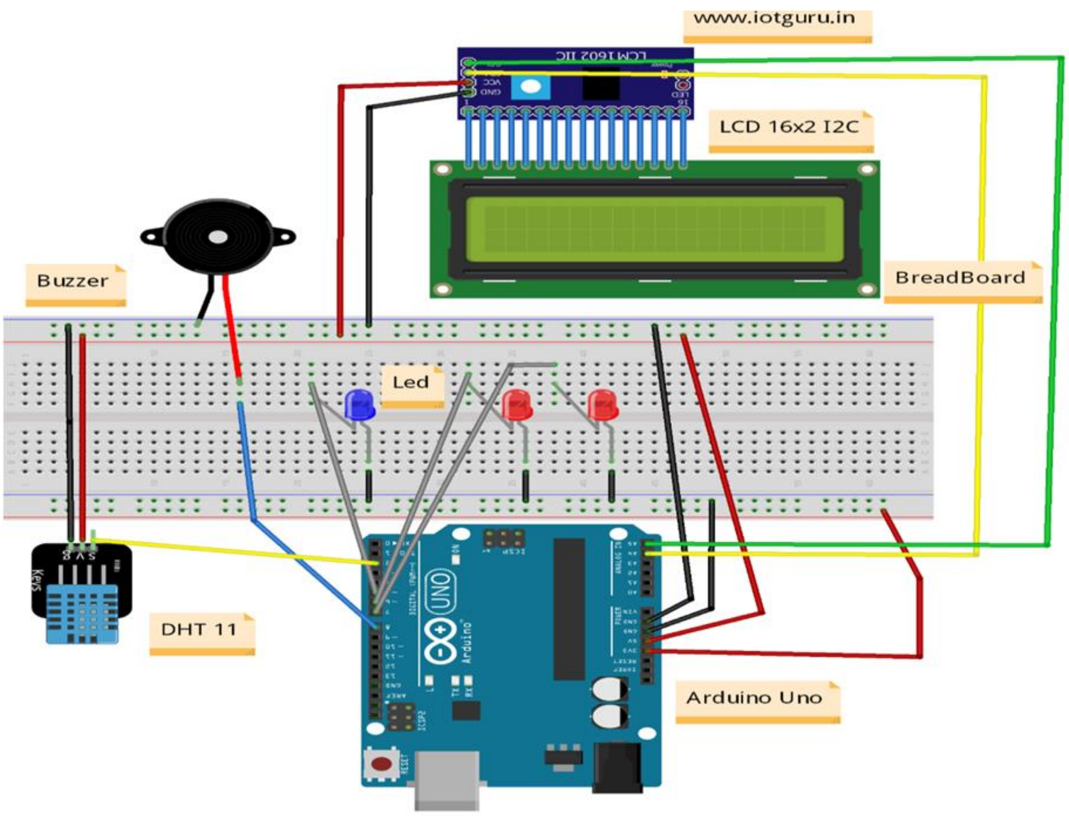
+ve leg to pin 7

Buzzer to Arduino board:

+ve pin to pin 8

-ve pin to gnd

Circuit diagram:



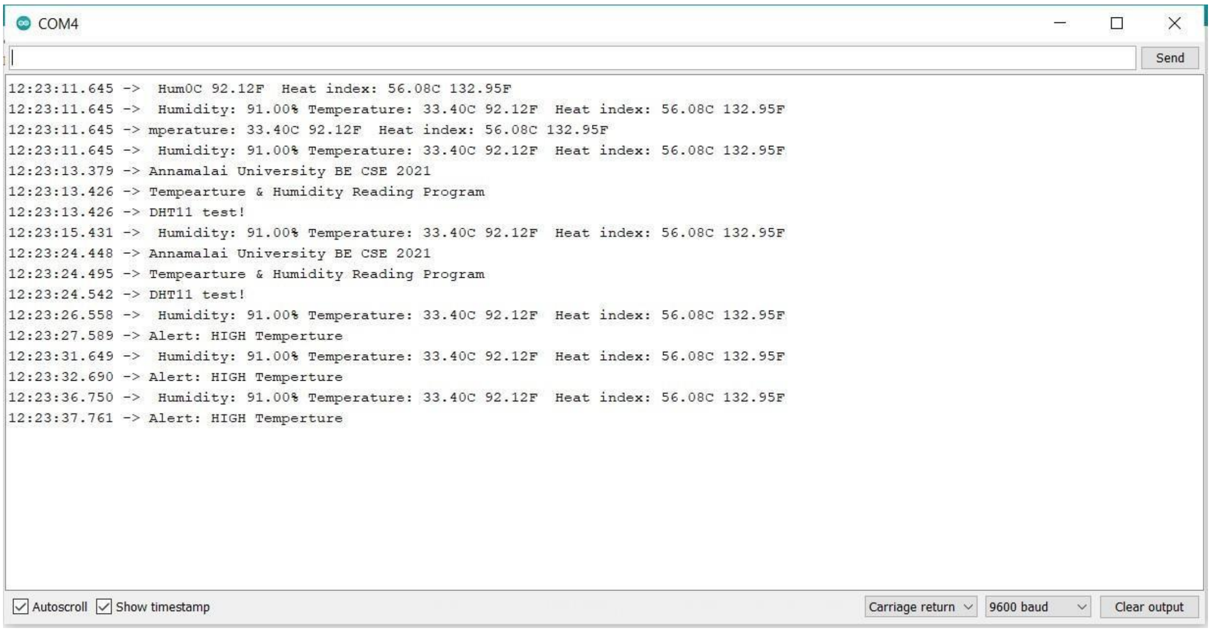
Code:

```
#include <Wire.h>
#include "DHT.h"
#define DHTPIN 2
#define DHTTYPE DHT11   // DHT 11
DHT dht(DHTPIN, DHTTYPE);
const int buzzer = 8;
const int light = 7;
const int light1 = 6;
const int light2 = 5;
void setup()
{ Serial.begin(9600);
  pinMode(buzzer, OUTPUT);
  pinMode(light, OUTPUT);
  pinMode(light1, OUTPUT);
  pinMode(light2, OUTPUT);
  Serial.println("Annamalai University BE CSE 2021 ");
  Serial.println("Tempearture & Humidity Reading Program");
  Serial.println(F("DHT11 test!"));
  dht.begin();
}
```

```
void loop()
  { digitalWrite(light, LOW);
   delay(2000);

  float h = dht.readHumidity();
  float t = dht.readTemperature();
  float f = dht.readTemperature(true);
  if (isnan(h) || isnan(t) || isnan(f))
    { Serial.println(F("Failed to read from DHT
    sensor!"));digitalWrite(light, HIGH);
    return; }
  float hif = dht.computeHeatIndex(f, h);
  float hic = dht.computeHeatIndex(t, h, false);
  Serial.print(F(" Humidity: "));
  Serial.print(h);
  Serial.print(F("% Temperature: "));
  Serial.print(t);
  Serial.print(F("C "));
  Serial.print(f);
  Serial.print(F("F  Heat index: "));
```

```
  Serial.print(hic);
  Serial.print(F("C "));
  Serial.print(hif);
  Serial.println(F("F"));
  if (t>29)
  { tone(buzzer, 1000);
  delay(1000);
  digitalWrite(light1, HIGH);
   digitalWrite(light2, LOW);
   tone(buzzer, 500); noTone(buzzer);
   Serial.println("Alert: HIGH Temperture");
    } else{
    Serial.println("Alert: Low Temperture");
    digitalWrite(light2, HIGH);
   }
  delay(2000);
  digitalWrite(light1,LOW);
}
```

Output:



Result:

Thus the room Temperature and Humidity was measured successfully using DHT11 sensor.

EX. NO: 05	COLOUR RECOGNITION USING ARDUINO
DATE :	

Aim:

To identify the different colours using colour recognition sensor.

List of components

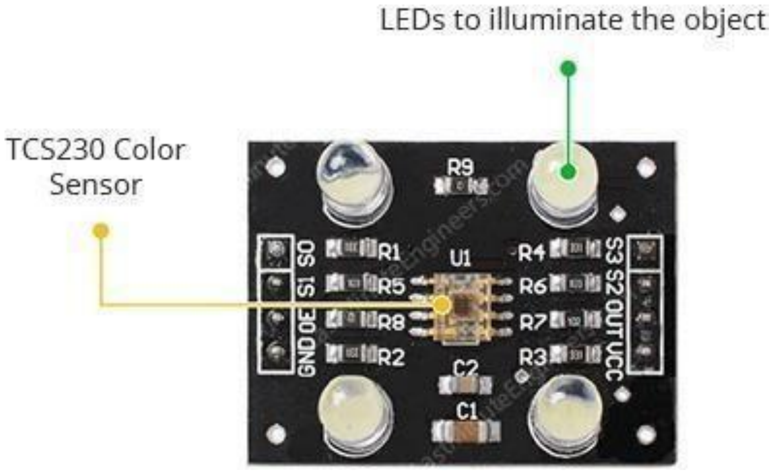
1. Jumper Wires
2. Arduino UNO
3. Colour recognition Sensor

Working Description

Colour sensors provide more reliable solutions to complex automation challenges. They are used in various industries including the food and beverage, automotive and manufacturing industries for purposes such as detecting material, detecting colour marks on parts, verifying steps in the manufacturing process and so on.

Working Principle of Colour recognition Sensor

The TCS230 colour sensor (also branded as the TCS3200) is quite popular, inexpensive and easy to use. At the heart of the module is an inexpensive RGB sensor chip from Texas Advanced Optoelectronic Solutions – TCS230. The TCS230 Colour Sensor is a complete colour detector that can detect and measure an almost infinite range of visible colours.



The sensor itself can be seen at the centre of the module, surrounded by the four white LEDs. The LEDs light up when the module is powered up and are used to illuminate the object being sensed. Due to these LEDs, the sensor can also work in complete darkness to determine the colour or brightness of the object. The TCS230 operates on a supply voltage of 2.7 to 5.5 volts and provides TTL logic-level outputs.

The TCS230 detects colour with the help of an 8 x 8 array of photodiodes, of which sixteen photodiodes have red filters, 16 photodiodes have green filters, 16 photodiodes have blue filters, and remaining 16 photodiodes are clear with no filters.

Connections:

Colour recognition sensor to Arduino board:

Vcc to 5V

Gnd to gnd

Pin S0 to pin 8

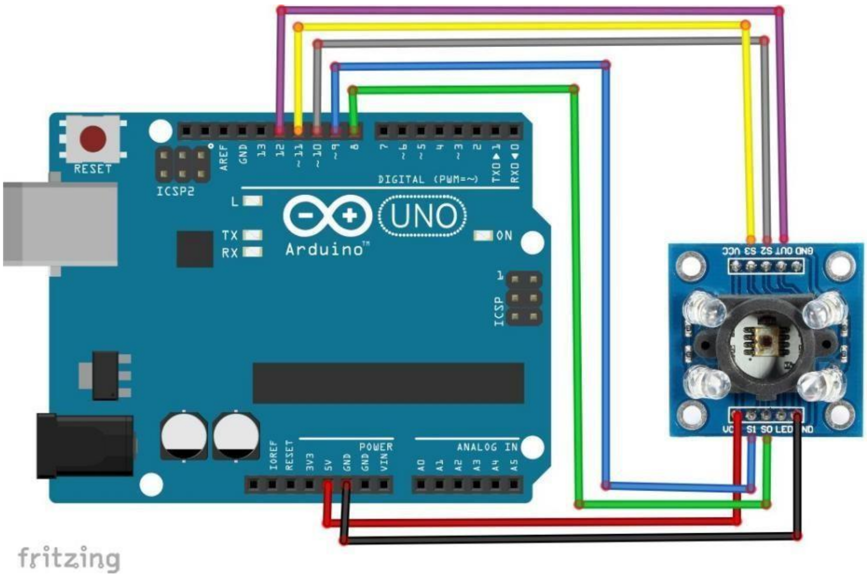
Pin S1 to pin 9

Pin S2 to pin 10

Pin S3 to pin 11

Out to pin 12

Circuit diagram:



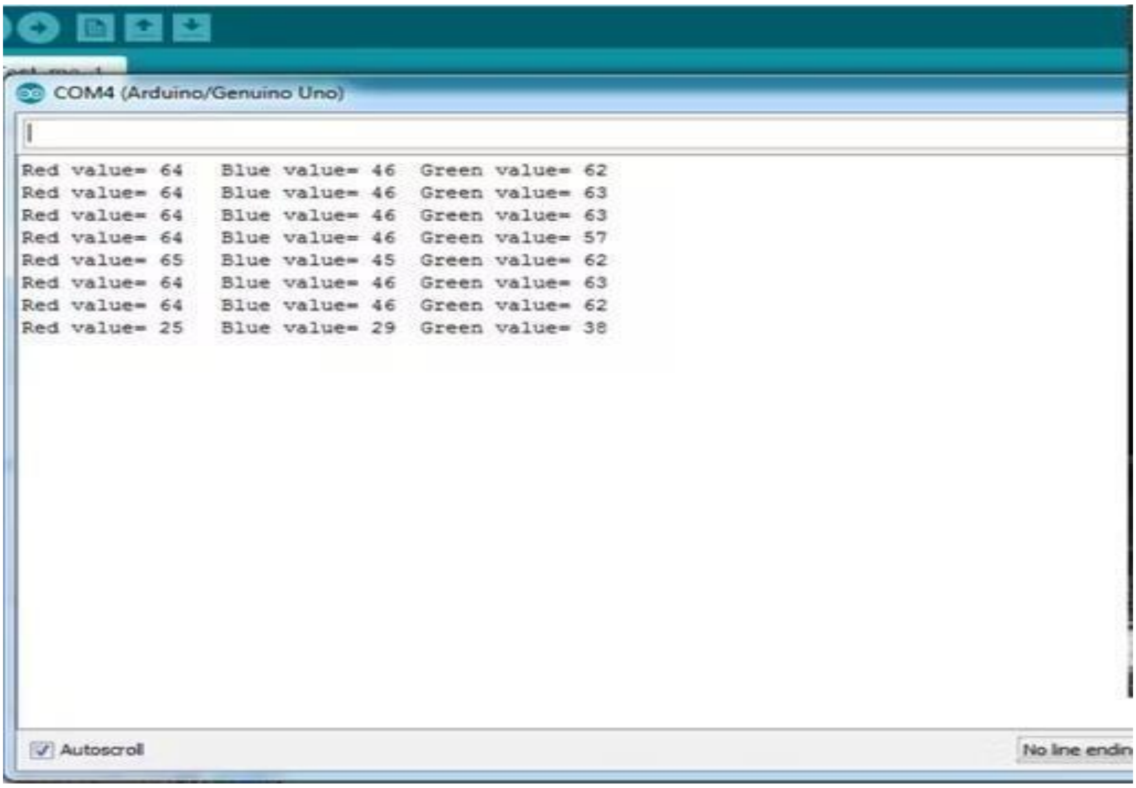
Code:

```
#define s0 8

#define s1 9
#define s2 10
#define s3 11
#define out 12
int data=0;

void setup()
{
  pinMode(s0,OUTPUT);

  pinMode(s1,OUTPUT);
  pinMode(s2,OUTPUT);
  pinMode(s3,OUTPUT);
  pinMode(out,INPUT);
  Serial.begin(9600);
```

```
  digitalWrite(s0,HIGH);
  digitalWrite(s1,HIGH);
  }
void loop()
{
  digitalWrite(s2,LOW);
  digitalWrite(s3,LOW);
  Serial.print("Red value= ");
  GetData();
  digitalWrite(s2,LOW);
  digitalWrite(s3,HIGH);
  Serial.print("Blue value= ");
  GetData();
  digitalWrite(s2,HIGH);
  digitalWrite(s3,HIGH);
  Serial.print("Green value= ");
  GetData();
  Serial.println();
  delay(2000);
}
```

```
void GetData(){
  data=pulseIn(out,LOW);
  Serial.print(data);
  Serial.print("\t");
  delay(20);
```

Output:



Result:

Thus, the colour recognition sensor has been interfaced with Arduino successfully.

EX. NO: 06	**FIRE ALARM INDICATOR USING ARDUINO**
DATE :	

Aim:

To develop a fire alarm indicator using the Flame sensor.

List of components:

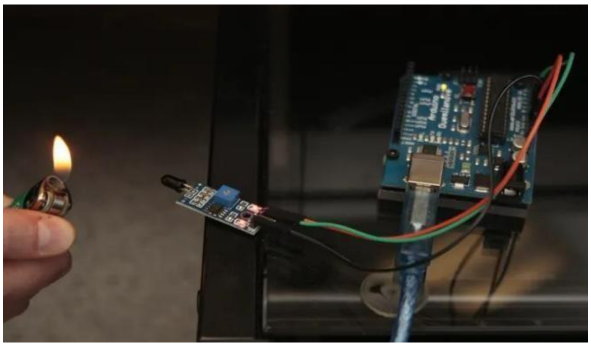
1. Solderless Breadboard Full Size
2. Jumper Wires
3. Arduino UNO
4. Buzzer
5. LEDs
6. Flame Sensor

Working Description:

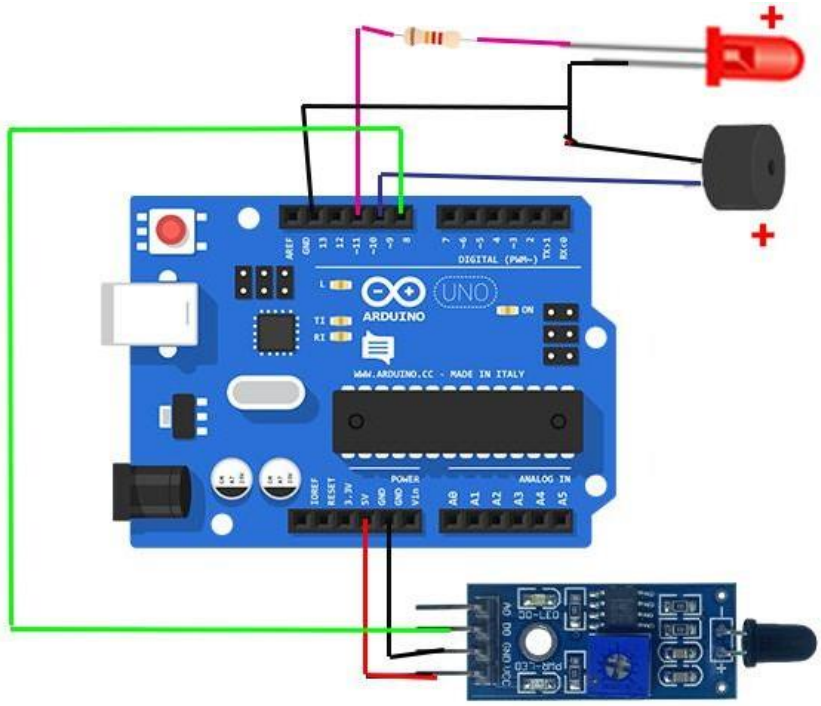
A flame-sensor is one kind of detector which is mainly designed for detecting as well as responding to the occurrence of a fire or flame. The flame detection response can depend on its fitting. It includes an alarm system, a natural gas line, propane & a fire suppression system. This sensor is used in industrial boilers. The main function of this is to give authentication whether the boiler is properly working or not. The response of these sensors is faster as well as more accurate compare with a heat/smoke detector because of its mechanism while detecting the flame.

Working Principle:

This sensor/detector can be built with an electronic circuit using a receiver like electromagnetic radiation. This sensor uses the infrared flame flash method, which allows the sensor to work through a coating of oil, dust, water vapour, otherwise ice.



Circuit diagram:



Connections:

Flame sensor to Arduino board:

Gnd to gnd

+ve pin to 5V

AO to A0

LED1 to Arduino board:

Gnd to gnd

+ve leg to pin 5

LED2 to Arduino board:

Gnd to gnd

+ve leg to pin 6

LED3 to Arduino board:

Gnd to gnd

+ve leg to pin 7

Buzzer to Arduino board:

+ve pin to pin 8

-ve pin to gnd

Code:

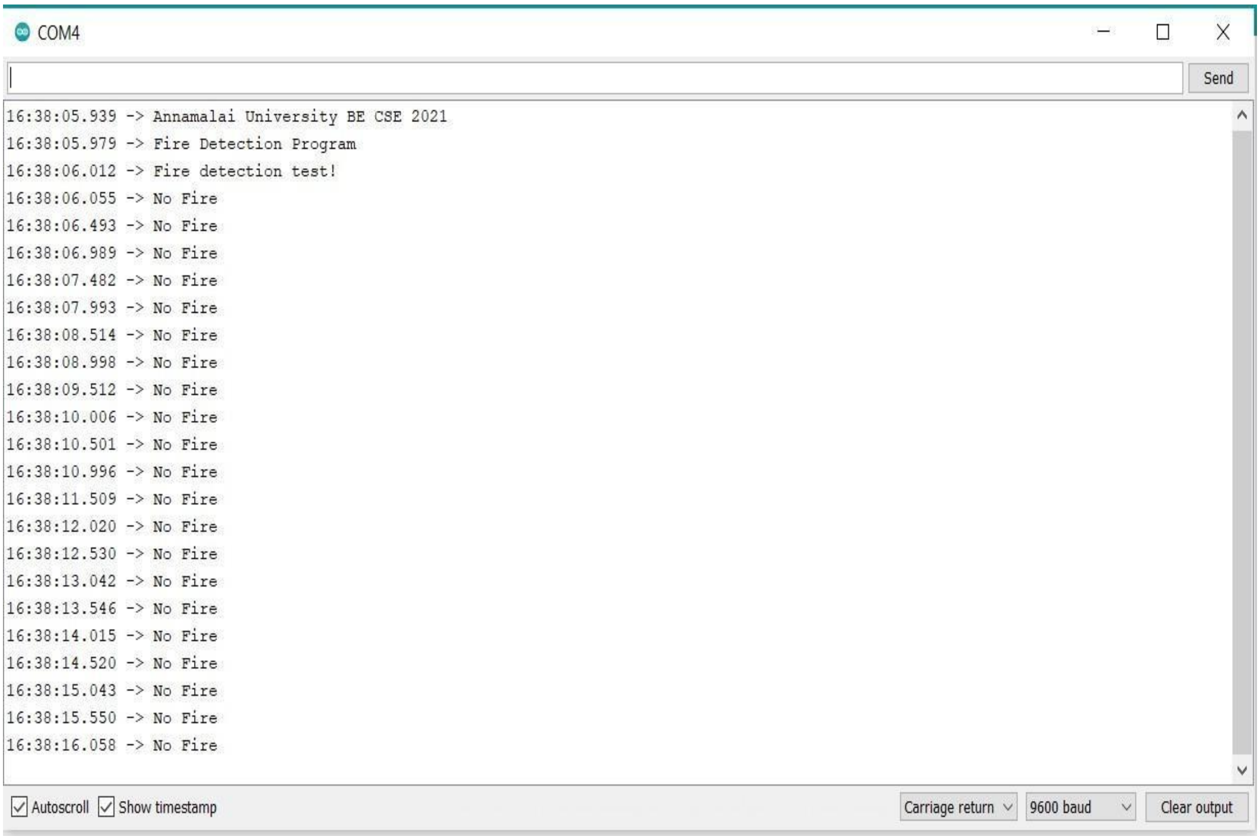
```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

const int sensorMin = 0;
const int sensorMax = 1024;
```

```
const int buzzer = 8;
const int light = 7;
const int light1 = 6;
const int light2 = 5;
void setup() {
Serial.begin(9600);
pinMode(buzzer, OUTPUT);
pinMode(light, OUTPUT);
pinMode(light1, OUTPUT);
pinMode(light2, OUTPUT);
  Serial.println("Annamalai University BE CSE 2021 ");
  Serial.println("Fire Detection Program");
  Serial.println(F("Fire detection test!"));
//    lcd.clear();
//    lcd.setCursor(0,0);
//    lcd.print("Starting the");
//    lcd.setCursor(0,1);
//    lcd.print("Fire Sensor");
}
```

```
void loop() {
  // read the sensor on analog A0:
 int sensorReading = analogRead(A0);
  // map the sensor range (four options):
  // ex: 'long int map(long int, long int, long int, long int, long int)'
  int range = map(sensorReading, sensorMin, sensorMax, 0, 3);
  // range value:
  switch (range) {

  case 0:    // A fire closer than 1.5 feet away.
    Serial.println("Close Fire");
//     lcd.clear();
//     lcd.setCursor(0,0);
//     lcd.print("***CLOSE***");
//     lcd.setCursor(0,1);
//     lcd.print("****FIRE****");
     tone(buzzer, 1000); // Send 1KHz sound signal...
     delay(1000);
     digitalWrite(light, HIGH);
    tone(buzzer,  500);
    noTone(buzzer);     // Stop sound...
    break;

 case 1:
 // A fire between 1-3 feet away.
    Serial.println("** Distant Fire **");
```

```
//     lcd.clear();
//     lcd.setCursor(0,0);
//     lcd.print("***DISTANT***");
//     lcd.setCursor(0,1);
//     lcd.print("****FIRE****");
    digitalWrite(light1, HIGH);
    break;
  case 2:
// No fire detected.
     Serial.println("No Fire");
//     lcd.clear();
//     lcd.setCursor(0,0);
 //     lcd.print("***NO***");
//     lcd.setCursor(0,1);
//     lcd.print("****FIRE****");
     digitalWrite(light2, HIGH);
     break;
  delay(500); // delay between reads
  digitalWrite(light, LOW);
  digitalWrite(light1, LOW);
  digitalWrite(light2, LOW);
}
```

Output:



Result:

Thus the fire alarm indicator was successfully implemented using flame sensor

EX. NO: 07	SOUND DETECTION USING ARDUINO
DATE :	

Aim:

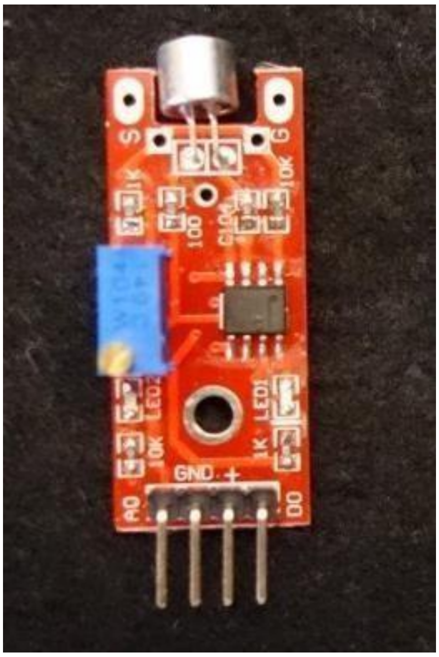
To develop a sound detection system using the sound sensor.

List of components:

1. Solderless Breadboard Full Size
2. Jumper Wires
3. Arduino UNO
4. LED
5. Sound Sensor

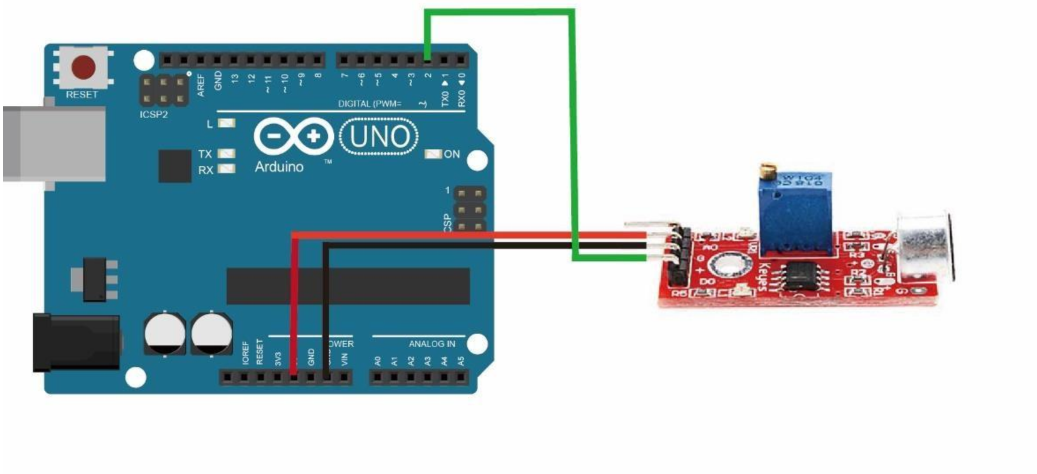
Working Description

The microphone sound sensor, as the name says, detects sound. It gives a measurement of how loud a sound is. There are a wide variety of these sensors. In the figure below you can see the most common used with the Arduino.



In this example, a microphone sensor will detect the sound intensity of your surroundings and will light up an LED if the sound intensity is above a certain threshold.

Circuit diagram:



Connections:

Sound sensor to Arduino board:

AO to A0

+ve pin to 5V

Gnd to gnd

LED to Arduino board:

+ve pin to pin 13

Gnd to gnd

Code:

```
const int ledPin = 13; //pin 13 built-in led
const int soundPin = A0; //sound sensor attach to A0
int threshold = 600; //Set minimum threshold for LED lit
void setup()
```

```
{
  pinMode(ledPin,OUTPUT);//set pin13 as OUTPUT
  Serial.begin(9600); //initialize serial
}
void loop()
{
  int value = analogRead(soundPin);//read the value of A0
  Serial.println(value);//print the value
  if(value > threshold) //if the value is greater than 600
  {
    digitalWrite(ledPin,HIGH);//turn on the led
    delay(200);//delay 200ms
  }
  else
  {
    digitalWrite(ledPin,LOW);//turn off the led
  }
  delay(1000);
```

Result:

Thus the sound detection system using Arduino was implemented successfully.

EX. NO:08	INTERFACING FLEX SENSOR WITH ARDUINO
DATE :	

Aim:

To interface a flex sensor with Arduino board.

List of components:

1. Solderless Breadboard Full Size
2. Jumper Wires
3. Arduino UNO
4. LED
5. Flex Sensor

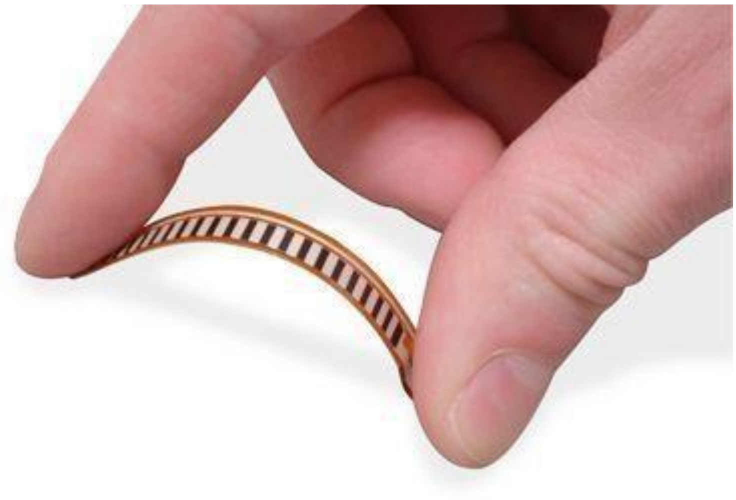
Working description:

A flex sensor is a kind of sensor which is used to measure the amount of defection otherwise bending. The carbon surface is arranged on a plastic strip as this strip is turned aside then the sensor's resistance will be changed. Thus, it is also named a bend sensor.

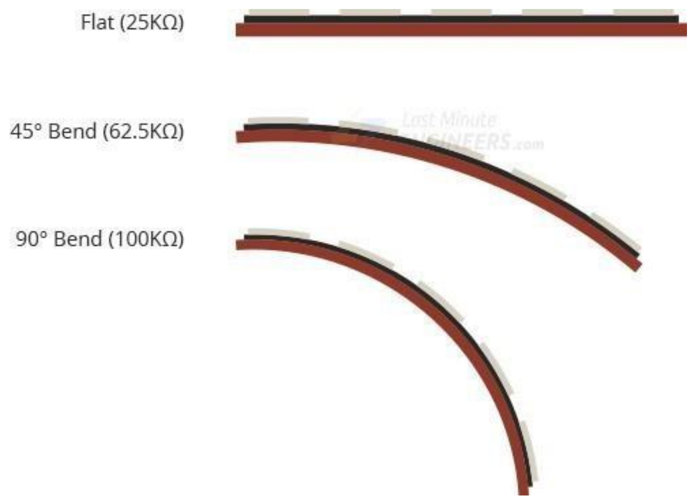
A flex-sensor could be used to check a door or window is opened or not. This sensor can be arranged at the edge of the door and once the door opens then this sensor also gets flexed. When the sensor bends then its parameters automatically change which can be designed to give an alert.

This sensor works on the bending strip principle which means whenever the strip is twisted then its resistance will be changed. This can be measured with the help of any controller. This sensor works similar to a variable resistance because when it twists then the resistance will be changed. The resistance change can depend on the linearity of the surface because the resistance will be dissimilar when it is level.

Flex sensors are designed to flex in only one direction – away from ink (as shown in the figure). Bending the sensor in another direction may damage it.

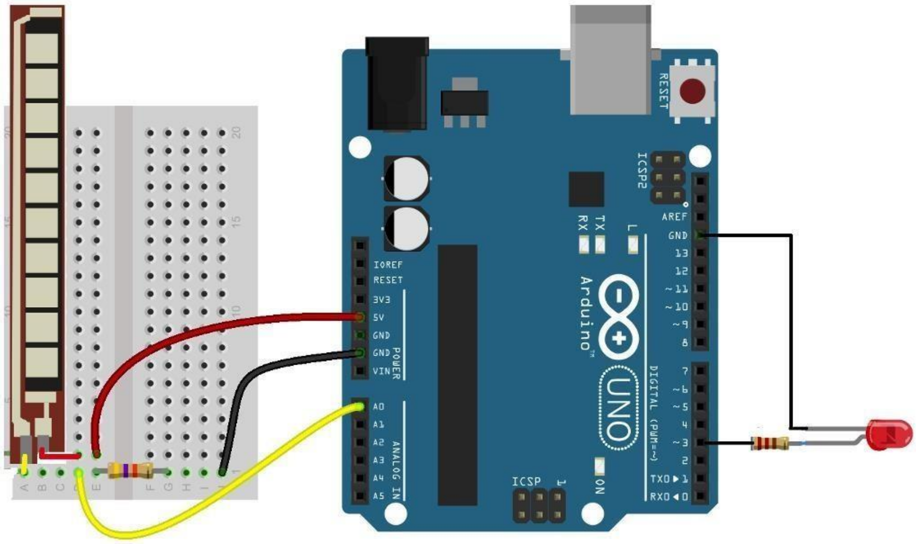


When the sensor is straight, this resistance is about 25k. When the sensor is bent, conductive layer is stretched, resulting in reduced cross section (imagine stretching a rubber band). This reduced cross section results in an increased resistance. At 90° angle, this resistance is about 100KΩ.



When the sensor is straightened again, the resistance returns to its original value. By measuring the resistance, we can determine how much the sensor is bent.

Circuit diagram:



Connections:

Flex sensor to Arduino board

VCC (pin 2) to 5 V

Pin 1 to 10 kΩ resistor to GND

Pin 1 to pin A0

LED TO Arduino board

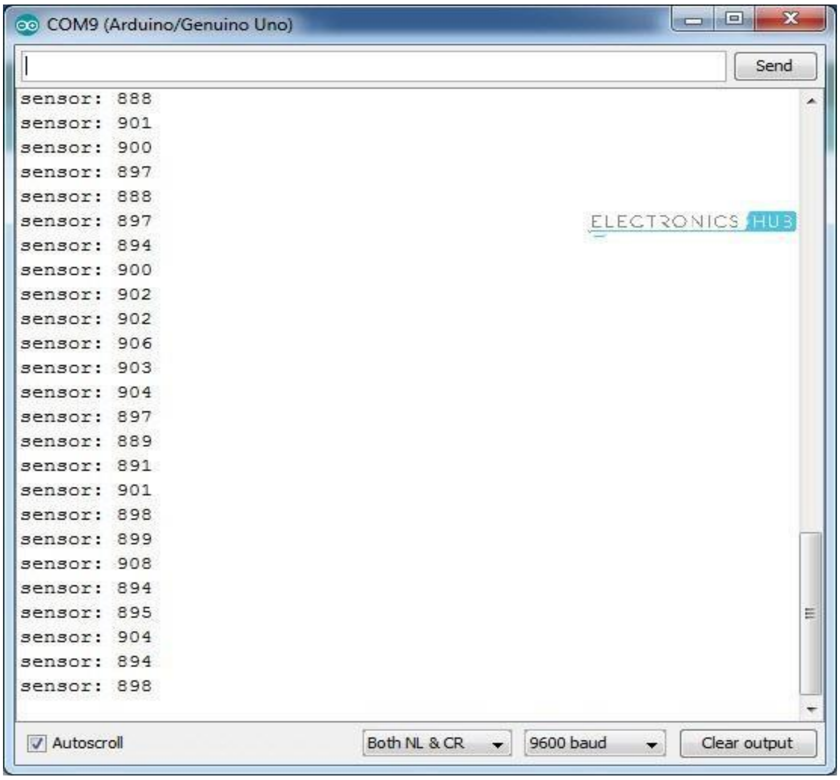
+ve pin to pin 3

-ve pin to GND

Code:

```
const int ledPin = 3;    //pin 3
const int flexPin = A0; //pin A0 to read analog input
int value; //save analog value
void setup(){
   pinMode(ledPin, OUTPUT); //Set pin 3 as 'output'
  Serial.begin(9600);       //Begin serial communication
}
void
loop(){ int
flexValue;
  flexValue = analogRead(flexPin);
  Serial.print("sensor: ");
  Serial.println(flexValue);
  if(flexValue>890)
     digitalWrite(ledPin,HIGH);
  else
    digitalWrite(ledPin,LOW);
    delay(1000);
}
```

OUTPUT:



Result:

Thus, the flex sensor was interfaced with Arduino successfully.

EX. NO:09	INTERFACING FORCE PRESSURE SENSOR WITH ARDUINO
DATE :	

Aim:

To interface a force pressure sensor with Arduino board.

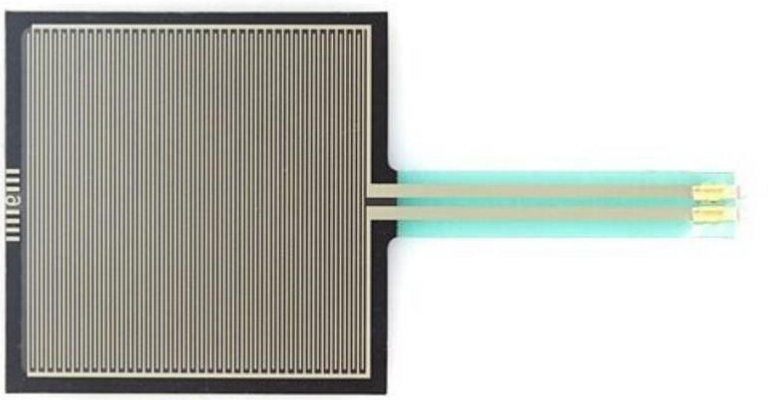
List of components:

1. Solderless Breadboard Full Size
2. Jumper Wires
3. Arduino UNO
4. Force pressure Sensor

Working description:

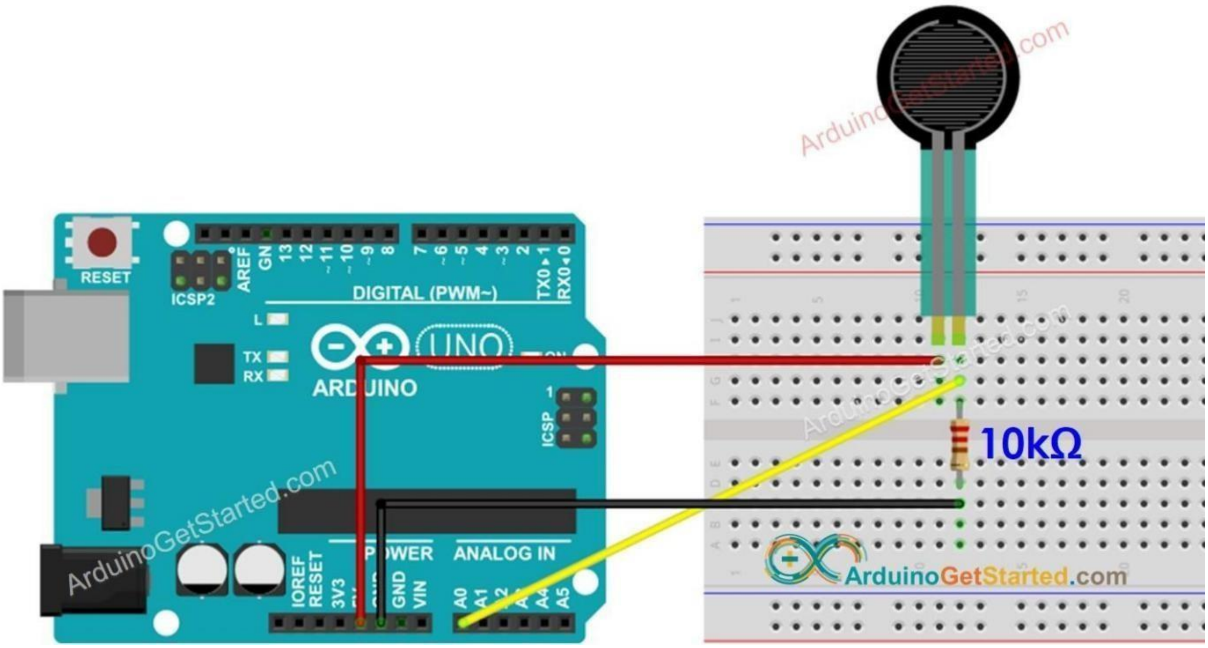
The working principle of a force sensor is that it responds to the applied force, as well as converts the value to a measurable quantity. Most force sensors are created with the use of force-sensing resistors. Such sensors consist of electrodes and sensing film.

Force-sensing resistors are based on contact resistance. These contain a conductive polymer film, which changes its resistance in a predictable way once force is applied on the surface.



In road traffic, force measurement plays an essential role. For instance, force sensors are being used in trucks. This means that the axle load may be determined precisely to enable effective and fast monitoring. Different force sensors are in use in automobiles. For instance, force sensors in the area of trailer couplings offer the possibility to know the trailer's load, and to determine static information in relation to dynamic driving behaviour on the road.

Circuit diagram:



Connections:

Flex sensor to Arduino board

VCC (pin 1) to 5 V

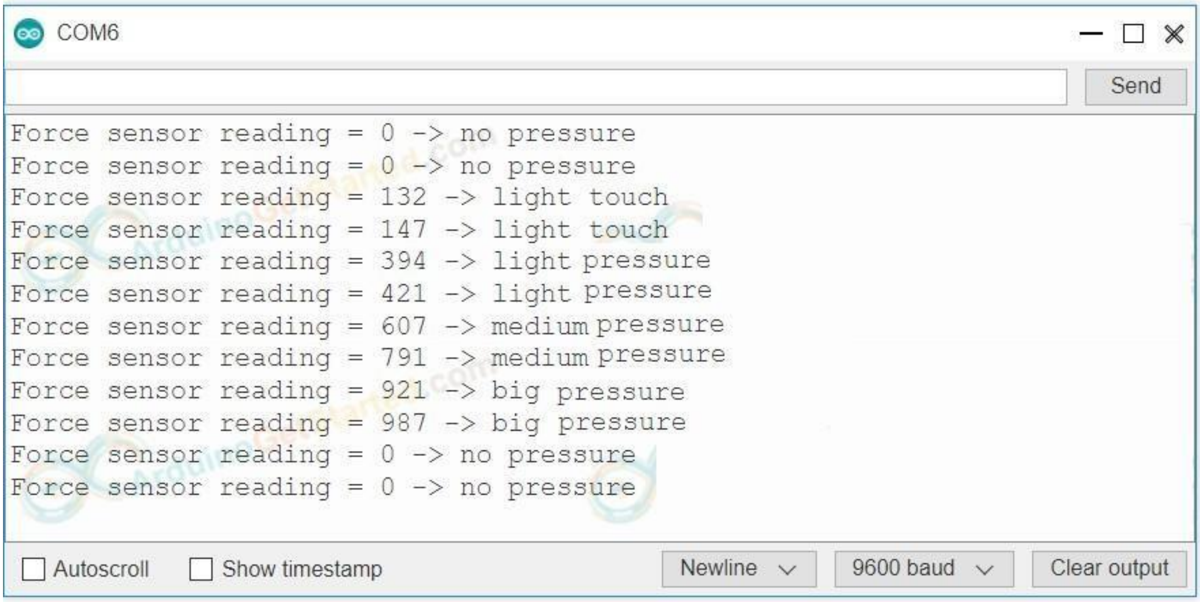
Pin 2 to 10 kΩ resistor to GND

Pin 2 to pin A0

Code:

```
#define FORCE_SENSOR_PIN A0
void setup() {
  Serial.begin(9600);
}
void loop() {
  int analogReading = analogRead(FORCE_SENSOR_PIN);
  Serial.print("Force sensor reading = ");
  Serial.print(analogReading); // print the raw analog reading
  if (analogReading < 10)       // from 0 to 9
    Serial.println(" -> no pressure");
  else if (analogReading < 200) // from 10 to 199
    Serial.println(" -> light touch");
  else if (analogReading < 500) // from 200 to 499
    Serial.println(" -> light pressure");
  else if (analogReading < 800) // from 500 to 799
    Serial.println(" -> medium pressure");
  else // from 800 to 1023
  Serial.println(" -> big pressure");
  delay(1000);
}
```

OUTPUT:



Result:

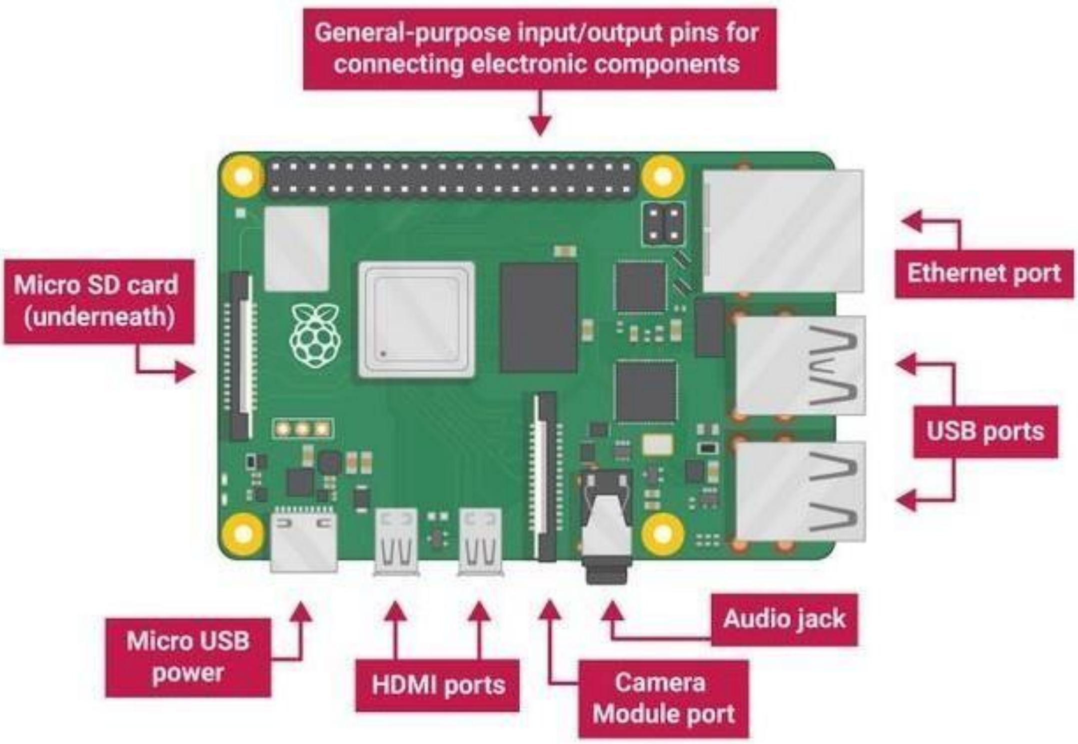
Thus the force pressure sensor was interfaced with Arduino successfully

Raspberry Pi

Introduction:

The Raspberry Pi is a low cost, **credit-card sized computer** that plugs into a computer monitor or TV, and uses a standard keyboard and mouse. It is a capable little device that enables people of all ages to explore computing, and to learn how to program in languages like Scratch and Python. It is capable of doing many tasks like browsing the internet and playing high-definition video, making spreadsheets, word-processing and playing games. The Raspberry Pi has the ability to interact with the outside world and has been used in a wide array of digital maker projects, from music machines and parent detectors to weather stations and tweeting birdhouses with infra-red cameras.

Pin configuration:



- **USB ports** — these are used to connect a mouse and keyboard. We can also connect other components, such as a USB drive.
- **SD card slot** — we can slot the SD card in here. This is where the operating system software and files are stored.

- **Ethernet port** — this is used to connect Raspberry Pi to a network with a cable. Raspberry Pi can also connect to a network via wireless LAN.
- **Audio jack** — we can connect headphones or speakers here.
- **HDMI port** — this is where we connect the monitor (or projector) that you are using to display the output from the Raspberry Pi. If the monitor has speakers, we can also use them to hear sound.
- **Micro USB power connector** — this is where we connect a power supply. We should always do this last, after we have connected all our other components.
- **GPIO ports** — these allow us to connect electronic components such as LEDs and buttons to Raspberry Pi.

Ex. No: 10	IDENTIFYING ROOM TEMPERATURE AND HUMIDITY USING RASPBERRY PI
DATE :	

Aim:

To identify the Room Temperature and Humidity using Raspberry Pi.

List of components:

1. Raspberry Pi
2. DHT 11 Sensor
3. Thingspeak cloud

Working Description:

1. Here, we are going to interface DHT11 sensor with Raspberry Pi 3 and display Humidity and Temperature on terminal.
2. We will be using the DHT Sensor Python library by Adafruit from GitHub. The Adafruit Python DHT Sensor library is created to read the Humidity and Temperature on raspberry Pi or Beaglebone Black. It is developed for DHT series sensors like DHT11, DHT22 or AM2302.
3. Extract the library and install it in the same root directory of downloaded library by executing following command

sudo python setup.py install

4. Once the library and its dependencies has been installed, open the example sketch named simple test from the library kept in examples folder.
5. In this code, raspberry Pi reads Humidity and Temperature from DHT11 sensor and prints them on terminal. But, it read and display the value only once. So, the program is developed in such a way to print value continuously.

Note:

Assign proper sensor type to the sensor variable in this library. Here, we are using DHT11 sensor.

sensor = Adafruit_DHT.DHT11

6. If anyone is using sensor DHT22 then we need to assign Adafruit_DHT.DHT22 to the sensor variable shown above.
7. Also, comment out Beaglebone pin definition and uncomment pin declaration for Raspberry Pi.
8. Then assign pin no. to which DHT sensor's data pin is connected. Here, data out of DHT11 sensor is connected to GPIO4.
9. Next this commands have to be executed in cmd

   sudo apt-get update

   sudo apt-get install build-essential python-dev python-openssl git

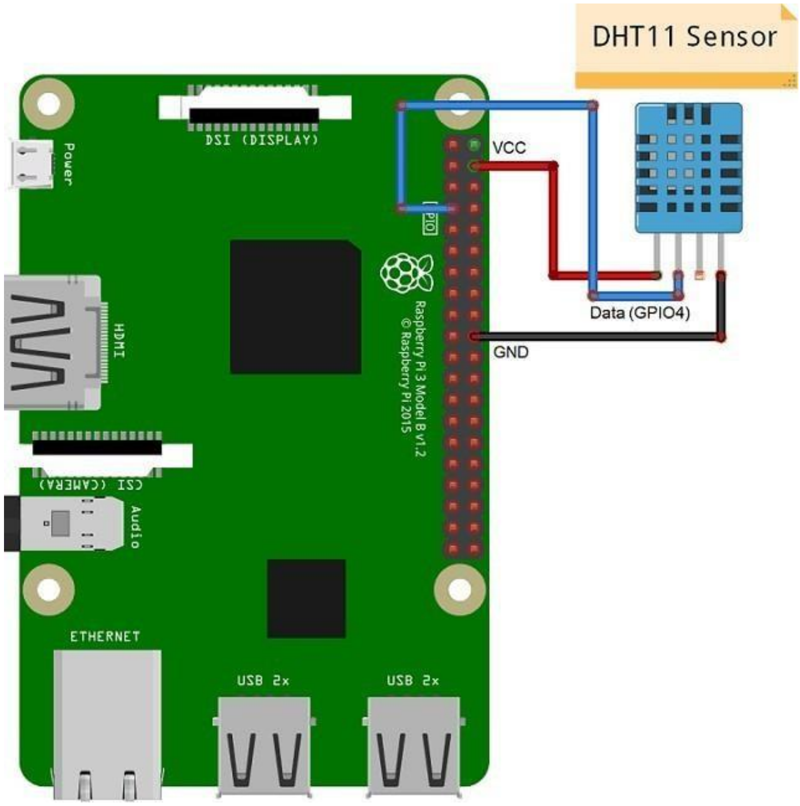
   git clone https://github.com/adafruit/Adafruit_Python_DHT.git && cd Adafruit_Python_DHT

   sudo python setup.py install.

10. Using Raspberry Pi ThingSpreak Library
11. In order to be able to use the service, it is possible to simply send the data via "POST" or retrieve via "GET". Functions are available in just about any programming language and with a little bit of knowledge, data transfer should be fast. The answers are in principle in JSON.
12. Alternative command

    sudo pip install thingspeak

Circuit diagram:



Connections:

DHT11 sensor to Raspberry Pi

Vcc to pin 4 (5 V)

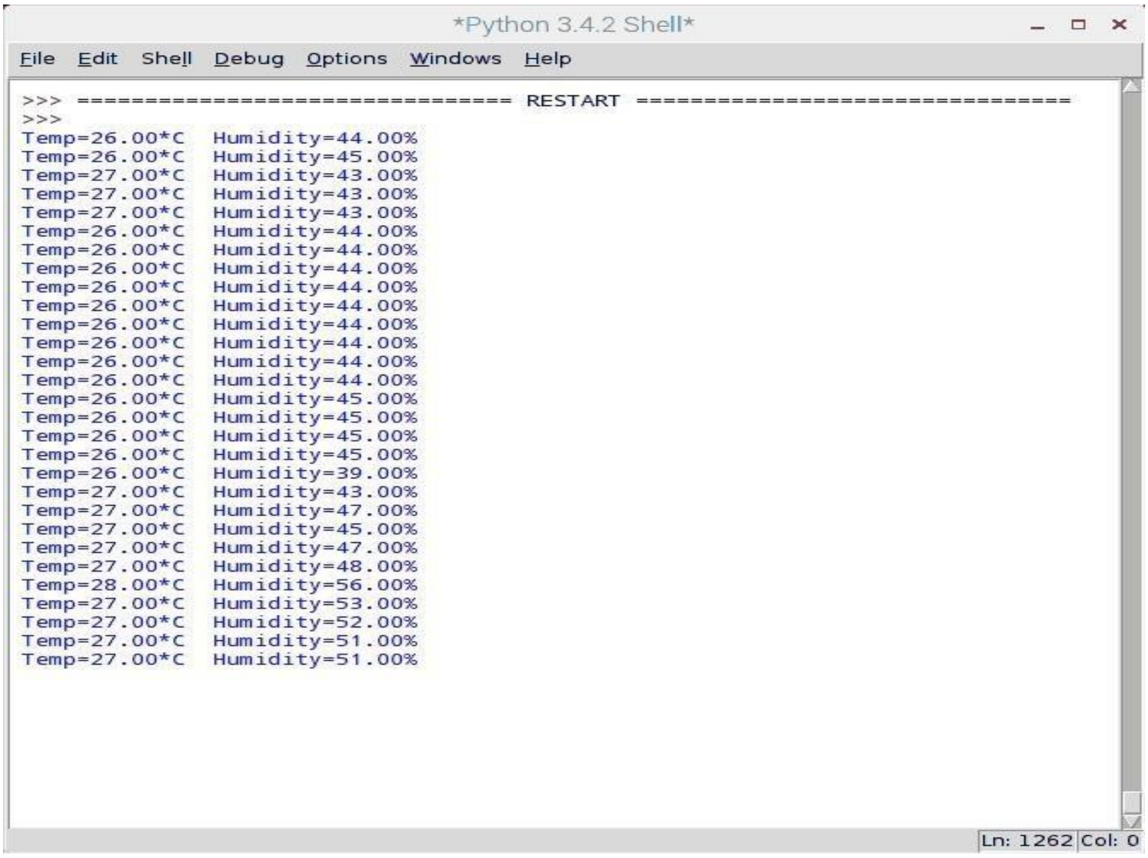
GND to GND

DATA pin to pin 7

Code:

```
import time
import adafruit_dht
import board

# Use GPIO 4 (pin 7)
dht_device = adafruit_dht.DHT11(board.D11)

while True:
    try:
        temperature = dht_device.temperature
        humidity = dht_device.humidity
        print(f"Temp: {temperature}°C    Humidity: {humidity}%")
    except RuntimeError as error:
        print(error.args[0])
    time.sleep(2)
```

Output:



```
*Python 3.4.2 Shell*
File  Edit  Shell  Debug  Options  Windows  Help
>>> ================================ RESTART ================================
>>>
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=45.00%
Temp=27.00*C  Humidity=43.00%
Temp=27.00*C  Humidity=43.00%
Temp=27.00*C  Humidity=43.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=44.00%
Temp=26.00*C  Humidity=45.00%
Temp=26.00*C  Humidity=45.00%
Temp=26.00*C  Humidity=45.00%
Temp=26.00*C  Humidity=45.00%
Temp=26.00*C  Humidity=39.00%
Temp=27.00*C  Humidity=43.00%
Temp=27.00*C  Humidity=47.00%
Temp=27.00*C  Humidity=45.00%
Temp=27.00*C  Humidity=47.00%
Temp=27.00*C  Humidity=48.00%
Temp=28.00*C  Humidity=56.00%
Temp=27.00*C  Humidity=53.00%
Temp=27.00*C  Humidity=52.00%
Temp=27.00*C  Humidity=51.00%
Temp=27.00*C  Humidity=51.00%
Ln: 1262 Col: 0
```

Result:

Thus the Room temperature and Humidity was measured using Raspberry Pi successfully.

EX. NO: 11	PIR MOTION SENSOR INTERFACING WITH RASPBERRY PI
DATE :	

Aim:

To build motion detection system using PIR sensor.

List of components:

1. Raspberry Pi
2. PIR Sensor
3. Jumper cable

Working Description:

PIR sensor is used for detecting infrared heat radiations. This makes them useful in the detection of moving living objects that emit infrared heat radiations. The output (in terms of voltage) of PIR sensor is high when it senses motion; whereas it is low when there is no motion (stationary object or no object).

PIR sensors are used in many applications like for room light control using human detection, human motion detection for security purpose at home, etc.

Setup:

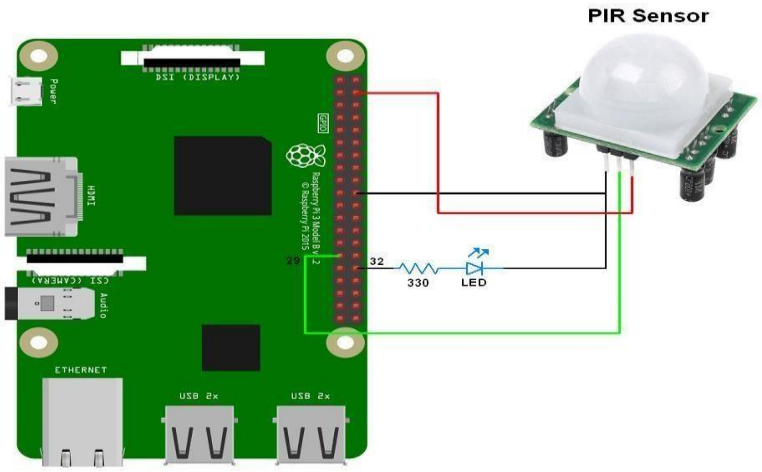
When motion is detected, PIR output goes HIGH which will be read by Raspberry Pi. So, we will turn on LED when motion is detected by PIR sensor. Here, LED is connected to GPIO12 (pin no. 32) whereas PIR output is connected to GPIO5 (pin no. 29).

Code:

```
import time
import RPi.GPIO as GPIO

# Set up the GPIO mode and PIR sensor pin
PIR_pin = 17  # GPIO pin 17 where the PIR sensor is connected BOARD PIN 11

# Set up GPIO using BCM numbering
```

```
GPIO.setwarnings(False)
GPIO.setmode(GPIO.BCM)
GPIO.setup(PIR_pin, GPIO.IN)  # Set the PIR pin as an input

print("Motion Detection Started")

try:
    while True:
        if GPIO.input(PIR_pin):  # If motion is detected (HIGH signal)
            print("Motion detected!")
        else:  # If no motion (LOW signal)
            print("No motion detected.")

        time.sleep(1)  # Wait for 1 second before checking again

except KeyboardInterrupt:
    print("Program interrupted by user.")

finally:
    GPIO.cleanup()  # Clean up GPIO settings when done
```

Circuit diagram:



Connections:

PIR sensor to Raspberry Pi:

VCC to VCC
GND to GND
DATA pin 11

Output:

Motion not detected..

Motion detected..

Motion not detected..

Motion not detected..

Result:

Thus the motion detector was built successfully using PIR sensor.

EX. NO:12	SOUND SENSOR INTERFACING WITH RASPBERRY PI
DATE :	

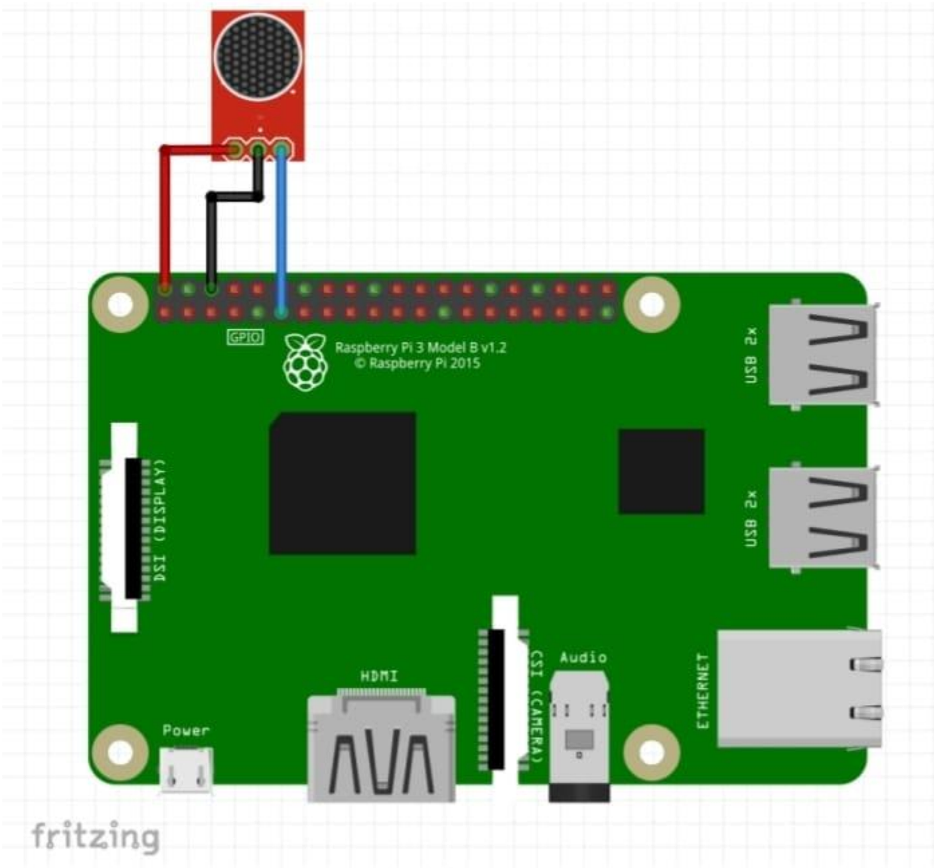
Aim:

To Interfacing sound sensor with Raspberry Pi

List of components:

1. Jumper Wires
2. Raspberry Pi
3. Sound sensor

Circuit diagram:



Connections:

Sound sensor to Raspberry Pi

D0 to pin 11
VCC to VCC
GND to GND

LED to Raspberry Pi

+ve pin to pin 3
-ve pin to gnd

Code:

```
from time import sleep
import RPi.GPIO as GPIO
GPIO.setwarnings(False)
GPIO.setmode(GPIO.BOARD)
GPIO.setup(11,GPIO.IN)
GPIO.setup(3,GPIO.OUT)
while True:
    if(GPIO.input(11)== True):
        GPIO.output(3,False)
        print("NO sound");
        sleep(1)
    if(GPIO.input(11)== False):
        GPIO.output(3,True)
        print("sound");
        sleep(1)
```

Output:

No Sound
No Sound
Sound
Sound

Result:

Thus the Sound Sensor interfacing using Raspberry Pi successfully.

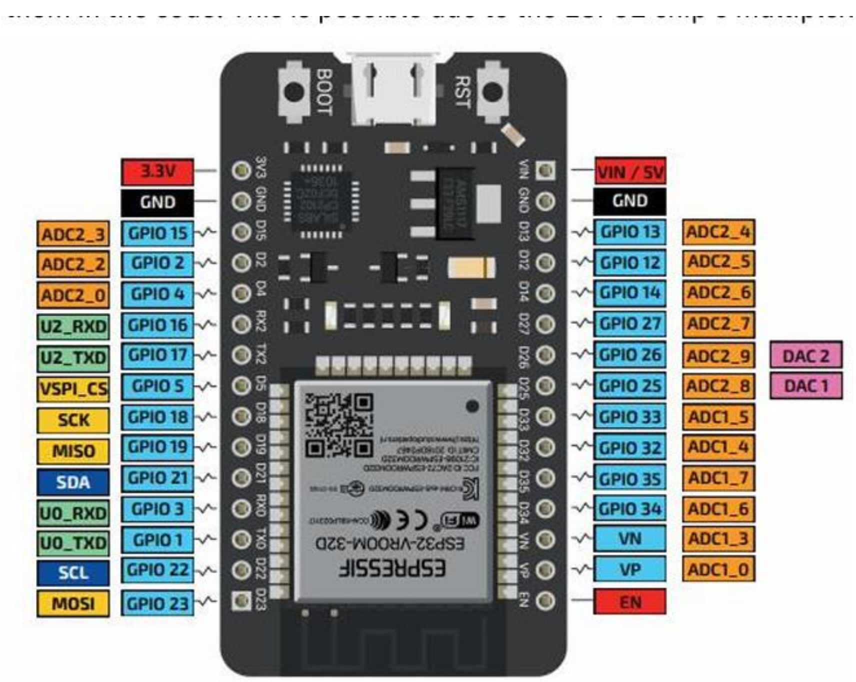
Introduction of ESP32

The ESP32 is a dual-core system with two Harvard Architecture Xtensa LX6 CPUs. All embedded memory, external memory and peripherals are located on the data bus and/or the instruction bus of these CPUs. ESP32 can perform as a complete standalone system or as a slave device to a host MCU, reducing communication stack overhead on the main application processor. ESP32 can interface with other systems to provide Wi-Fi and Bluetooth functionality through its SPI / SDIO or I2C / UART interfaces. The two CPUs are named “PRO_CPU” and “APP_CPU” (for “protocol” and “application”), however, for most purposes the two CPUs are interchangeable.

GPIO Pins: The ESP32 peripherals include:

- 18 Analog-to-Digital Converter (ADC) channels
- 3 SPI interfaces
- 3 UART interfaces
- 2 I2C interfaces
- 16 PWM output channels
- 2 Digital-to-Analog Converters (DAC)
- 2 I2S interfaces
- 10 Capacitive sensing GPIOs

The ADC (analog to digital converter) and DAC (digital to analog converter) features are assigned to specific static pins. However, you can decide which pins are UART, I2C, SPI, PWM, etc – you just need to assign them in the code. This is possible due to the ESP32 chip’s multiplexing feature.



EX. NO:13	ANALOG SENSOR WITH ESP32 MICROCONTROLLER
DATE:	

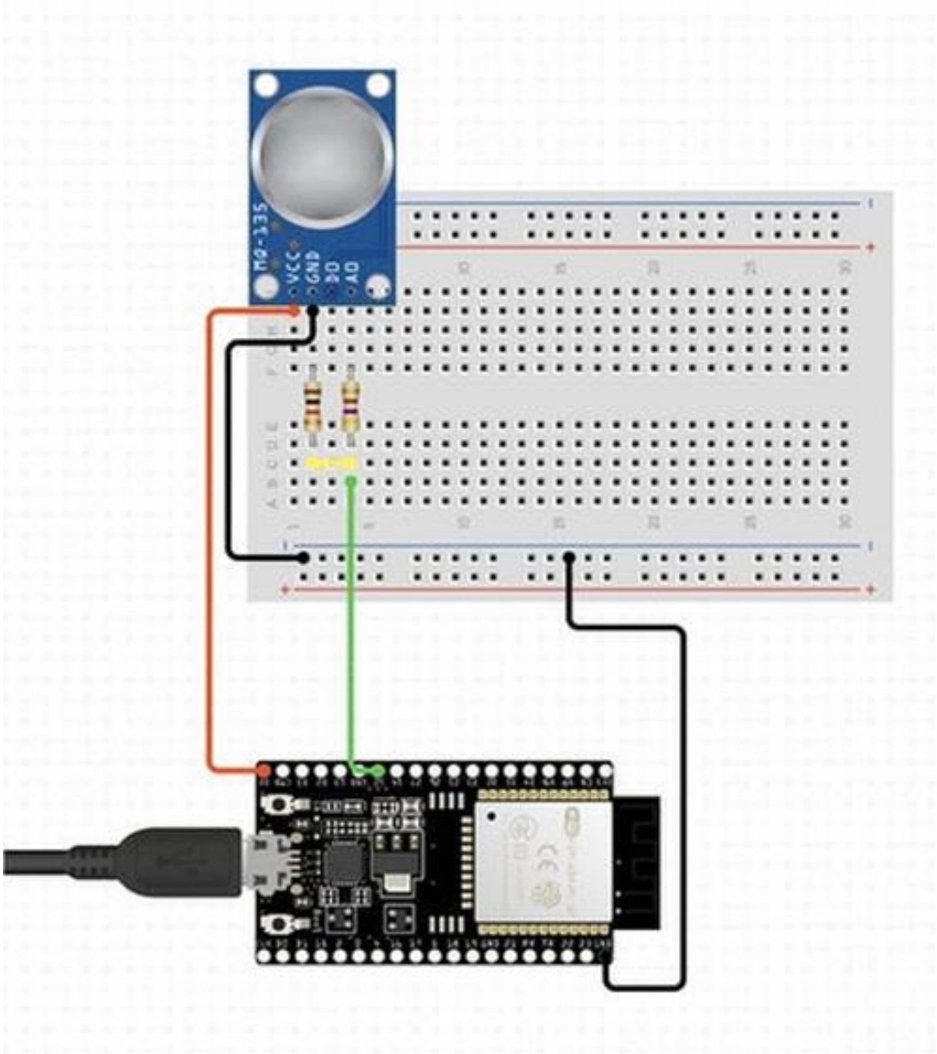
Aim:

To Interface an analog sensor (MQ135) with an ESP32 Microcontroller.

List of components:

1. **ESP32 Microcontroller**
2. **MQ135 gas sensor**
3. **Breadboard**
4. **Jumper wires**

Circuit diagram:



Work description:

Sensor (MQ135) Working Description The MQ135 is an analog gas sensor that detects air quality or the presence of certain gases in the air.

1. The MQ135 sensor is connected to GPIO pin 26 (analog input) of the ESP32.
2. The analogRead(26) function reads the analog output of the MQ135 sensor.
3. The sensor's analog output varies based on the concentration of gases detected.
4. The code prints this analog value to the Serial Monitor for monitoring or further processing.

Code:

```
void setup() {
     Serial.begin(9600);
     pinMode(26, INPUT);
    }

void loop() {
     int sensorValue = analogRead(26);
     Serial.println(sensorValue);
    delay(1);
    }
```

Output:

Result:

Thus the analog sensor show analog value from the ESP32 using successfully.

EX. NO:14	**DHT11 SENSOR WITH ESP 32**
DATE:	

Aim:

To interface DHT11 with ESP32 to read temperature and humidity.

List of components:

1. ESP32 Microcontroller
2. DHT11 sensor
3. Jumper wires

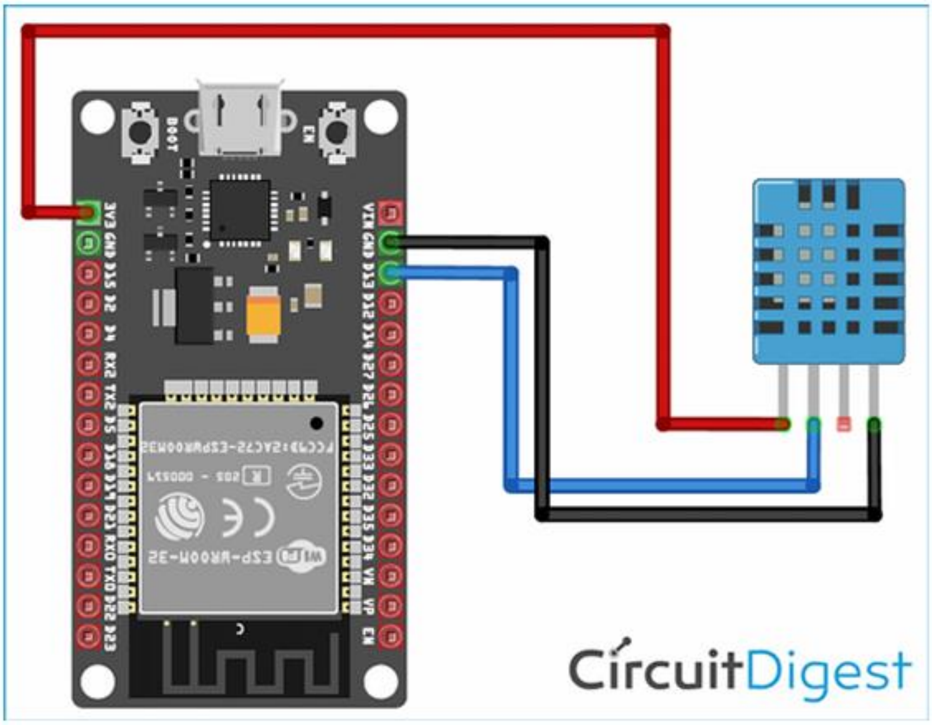
Working description:

The DHT11 sensor measures temperature and humidity. When connected to an ESP32, it sends digital signals representing the temperature and humidity values. The ESP32 reads these signals and prints the values to the Serial Monitor.

1. The DHT11 sensor captures temperature and humidity data.
2. The ESP32 reads the digital signal from the DHT11.
3. The ESP32 processes the data and prints it to the Serial Monitor.

This setup allows for monitoring environmental conditions using the DHT11 sensor with the ESP32.

Circuit diagram:



Code:

```
#include "DHT.h"
#define DHT11PIN 16 // Define the pin where DHT11 is connected
DHT dht(DHT11PIN, DHT11); // Initialize DHT11 sensor
void setup() {
  Serial.begin(115200);
  dht.begin(); // Start Serial communication // Start the DHT11 sensor
}
void loop() {
   float humi = dht.readHumidity(); // Read humidity
   float temp = dht.readTemperature(); // Read temperature in Celsius
   Serial.print("Temperature: ");
   Serial.print(temp);
   Serial.println(" ºC");
   Serial.print("Humidity: ");
   Serial.print(humi);
   Serial.println(" %");
   delay(1000); // Wait 1 second between readings
}
```

Output:

```
Reading from DHT11...
Temperature: 28.0°C
Humidity: 60.0%
------------------------
Temperature: 28.1°C
Humidity: 59.8%
```

Result:

Thus, the temperature and humidity readings from DHT11 sensor display in serial monitor successfully